



Vantage® Research Ultrasound System

Vantage System User Manual

D00023 Rev. A

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Introduction

The Vantage System User Manual provides important safety and regulatory guidelines for proper use of the system; detailed instructions on software installation, final setup, and verifying system functionality; information on backward compatibility constraints associated with certain software releases and system hardware configurations; an overview of the key system features; information on software capabilities that can be licensed, and on software license management; and references to relevant supporting documentation.

Verasonics recommends that the users thoroughly review this document before operating the Vantage System.

For information on the MATLAB release version and host controller OS version supported by the current Vantage software release, refer to Section 4.

1. Safety and Regulatory Guidelines

Users are required to adhere to the safety guidelines outlined in this section. If the system is not used in a manner specified by Verasonics, the protection provided by the equipment may be impaired.

1.1 Intended Use

The Vantage System is a laboratory research and/or development platform, and is intended to be used to acquire, store, display, and analyze ultrasound data; it is not intended for use on human subjects.

1.2 Safety Symbols and Description

The following safety labels are displayed on the Vantage hardware system chassis to identify any potential hazards and to ensure safe operation.

Table 1: Safety Symbols - Description

Safety Symbol	Description
	Refer to the Vantage System User Manual before operating the system.
	Caution This safety symbol indicates a potentially hazardous situation that could result in minor or moderate injury—refer to the Safety Guidelines section covered in this document.
	Hazardous Voltage This symbol indicates the presence of hazardous voltages inside the Vantage hardware system enclosure. Note that there are no user-serviceable components inside the chassis. <i>Do not remove the side covers of the hardware system.</i>

1.3 Safety Guidelines

The Vantage hardware system chassis houses high-voltage components that can produce emissions such as heat and acoustic output, which can potentially cause harm or injury. Users must follow the safety guidelines noted below.

Always operate the Vantage hardware system with its covers securely fastened in place.

 The covers provide system cooling, EMC shielding, and protection against hazardous voltages present on certain exposed parts inside the Vantage hardware system chassis.

Note: There are no user-serviceable parts within the Vantage hardware system—the system should be returned to Verasonics for all servicing needs such as hardware-level repairs and upgrades. *Any evidence of unauthorized removal of the side covers may result in the termination of the system warranty.*

 To protect the system and users from electrical hazards, ensure there are no liquid spills on or into the Vantage hardware system chassis. If a spill occurs, shut down the host controller and the Vantage hardware system immediately, and notify the [Verasonics Support team](#).

Note: Ensure that any measures taken to protect the system from spills do not restrict the cooling airflow.

Ensure that the Vantage hardware system is always stored and operated in its upright position.

- Using the Vantage hardware system in any other orientation—such as placing the system on its side—can result in hardware damage. *Any hardware damage caused due to operating the Vantage hardware system in an incorrect orientation is not covered under the warranty.*

Allow for system cooling.

- The Vantage System is intended for use as laboratory equipment; and should be placed on a bench/desk or on the floor with adequate spacing (at least six inches of spacing) around it to allow for adequate air circulation.
- If the system is placed on the floor, ensure that it is placed on a hard surface and not on a carpeted area.
- Note that it is required to provide adequate space behind the rear end of the system to avoid over-bending of the PCIe adapter cable.
- Do not store anything underneath the Vantage hardware system—this can inhibit airflow and cause the system to overheat.
- If the cooling fans do not operate when the Vantage hardware system and the host controller are on, please shut the system down and contact the [Verasonics Support team](#) immediately.

Conduct EMC tests on the overall system configuration when the system is used in an environment where RF interference is a concern.

-  Shield any third-party devices in the lab that may cause interference—custom transducers that are poorly shielded can interfere with other instrumentation. In the presence of medical devices sensitive to RF radiation, conduct EMC tests of the overall system configuration (transducer, Vantage hardware system, host controller, and any other connected devices).
- Note that conducting EMC tests on the individual components is not a meaningful way to predict the EMC performance of the overall system.
- Third-party (non-Verasonics) transducers might not adhere to the standards outlined in IEC 61010-1.
- System radiated emissions can be impacted by many factors including but not limited to transducer construction (grounding, shielding, and element tuning) and transmit settings (waveforms, pulse repetition frequencies, aperture sizes, delays, and voltage settings). Customized and/or modified example scripts may create conditions that exceed FCC and IEC limits.

Perform AC Line safety tests.

- An end-to-end Vantage hardware system is interconnected with other devices—such as a host controller and a display monitor—each of which must have its own AC line power connection.
-  The net AC line leakage current of these interconnected devices can exceed safe operating limits, and other errors in the AC line power connections (for example, inadequate ground connections) can lead to potential safety hazards.
- Conduct AC line safety tests for isolation, leakage current, and ground impedance across the entire interconnected system. Repeat AC line safety tests whenever the system is disassembled and reassembled or is subjected to rough transport or other environmental stresses.
- Ensure that the system operating environment complies with the following electrical and environmental requirements:

Table 2: Electrical and Environmental Specifications

Features	Description
AC Line Power Input	There is an IEC connector module located at the back of the system, for use with a detachable power cord. This module also provides an AC line disconnect switch and fuse. From this module, the AC line power is routed directly to the OEM 12 Volt power supply mounted within the system. The OEM supply provides universal AC line input capability: 100 – 240 V, 50 – 60 Hz.
Power Consumption Estimates	▪ 5 Watts or less: System in shutdown state (Vantage system is powered down, but the internal OEM AC power supply is in its standby operating state, ready to power up the system under the control of the host controller). ▪ 200 Watts: Vantage 256 system powered up and initialized, in an idle state with no transmit-receive activity. ▪ 280 Watts: Vantage 256 system in a typical operating state running an imaging script. ▪ 500 Watts: Vantage 256 system in a worst-case therapeutic HIFU transmit operating state (in external HIFU configuration-only mode), near-CW transmit activity with all 256 transmit channels active. Note that these power consumption estimates apply only to the Vantage hardware system and do not include power consumed by the host controller, display monitor, other peripheral devices, or the external HIFU transmit power supply in the case of the HIFU system configuration.
AC Line Isolation/Emission/Susceptibility	The OEM 12 Volt power supply provides a 'medical grade' AC line input capability in terms of leakage current and other regulatory requirements. The only devices in the Vantage hardware system with any connection or proximity to the AC line are the OEM power supply and the IEC connector module at the rear of the system. For details of the regulatory/safety characteristics of the AC line input, refer directly to the manufacturer's specifications and datasheets for these two devices. OEM Power Supply: TDK/Lambda CSS500-12 IEC Connector module: TE Connectivity PE0S0DHXB
Environmental Specifications	Description
Operating Temperature Range	▪ For Vantage-256 system configurations: 10°C to 30°C. (50 to 86 ° F.) ▪ For all other Vantage system configurations: 10°C to 35°C. (50 to 95° F)

Clean the exterior surface of the system using the following guidelines:

1. Remove any loose dust on the top, back, and sides of the system using a lint-free cloth.
2. Use a soft cloth dampened with water to clean the system.

Note: While cleaning the system, exercise caution to prevent any damage to the optical PCIe cable and port located at the back of the system. Do not use abrasive or chemical cleaning agents that might cause damage to the surface of the Vantage System.

Follow software installation instructions carefully.

- Follow instructions, especially with power cycling (cold reboot), to ensure proper software configuration when FPGA or firmware updates have been made.

Conduct a self-test before calibrating the acoustic output of the system.

- If the user intends to calibrate the acoustic output of the system, "VerasonicsVerificationTest" (or equivalently, "VVT") should be run before data is taken to confirm that the system is performing within specifications.
- Acoustic output measurements or any other verification tests done on the system should be repeated after every software or hardware modification, upgrade, or repair.

Do not place the Vantage System in a damp/wet area.**Power off the AC switch before unplugging the Vantage System.****Conduct adequate user-script verification.**

- The Vantage System is designed to be flexible and easily programmable for a wide variety of applications. An inherent risk with this flexibility is that a system programming error can result in unexpected system behavior or acoustic output than what was intended, with no warning or fault condition indication from the system.
- Similarly, in the definition of transmit waveforms and transmit beam steering/focusing, the user should conduct sufficient testing of the transmit waveform and acoustic output to ensure that the actual output waveform and power level match what was intended.

Evaluate audible noise levels.

- The cooling fans used in the Vantage System, in the host controller, and in any other devices that are included as a research and development system, can result in a cumulative audible noise level that can annoy or distract the operator.

Hardware-Software Handshaking and Fault Detection

When the Vantage System is running, the actual acquisition and processing functions are divided into two independent event sequences that are very loosely coupled with each other—one running within the Vantage hardware system, and the other in software on the host controller. This "loose coupling" creates a potential hazard condition (especially in situations where the system is used on human subjects) if one of the sequencers has malfunctioned and the other is not aware of it. A mitigation for this potential hazard (as identified in the [Vantage Risk Analysis](#) document) is to implement "watchdog functions" in both the hardware and software sequencers, which trigger a fault condition if the other sequencer does not respond to a periodic query within a user-programmable time limit.

- The Vantage System includes a software watchdog timer running in the Vantage hardware system that terminates execution of the hardware sequencer and triggers a system fault condition if the timer has not been reset by the end of the timeout interval. The timer is reset periodically by the Vantage software running on the host controller, and thus will not be reset if the software has locked up or crashed. This watchdog feature is always enabled whenever the system is executing a sequence. Refer to the [Vantage Sequence Programming Manual](#) for details on the software watchdog and on how to program the timeout interval.
- The Vantage System does not include an automatic watchdog going the other way, to allow the system software to determine whether the hardware sequence is functioning properly. The user can implement the "hardware watchdog" in the event sequence. The simplest way to do this is with a `sync` command, which includes a user-programmable timeout interval. If the software event sequence is waiting at a `sync` command point for the signal from the Vantage hardware system that it has also reached that point, and the sync timeout interval expires before that signal is received, the software reports an error condition and forces both sequencers to stop. Similar hardware watchdog handshaking can also be achieved through the use of synchronous DMA transfer ("`transferToHost`") commands, that have a user-programmable DMA timeout interval which can be used to shut down the system with a DMA timeout error condition. Note, however, that in some operating states, the DMA timeout may be disabled even though synchronous DMA transfers are being used (one such example is Recon processing of the RF data while using the -1 "most recent frame" RF data frame pointer). Refer to the

Vantage Sequence Programming Manual for details on how to use the `sync` command and synchronous DMA commands, and how to program their respective timeout intervals.

1.4 EMI Immunity

If obvious noise appears in the ultrasound image, there may be a radiated or conducted immunity issue. Contact the Verasonics Support team at support@verasonics.com.

1.5 Protective Grounding

 Upon installation of the Vantage System, ensure that the system is connected to an outlet that provides reliable protective grounding. When plugging into an electrical outlet, ensure that the outlet is properly grounded and rated for use with the Vantage NXT hardware system. In addition, ensure that the system is connected to the grounded outlet with the appropriate three-pronged (grounded) plug supplied by Verasonics.

1.6 Damage in Shipping and Handling

Carefully examine the shipping container for any signs of physical damage upon receiving a new Vantage System or when a Vantage System is shipped back after an upgrade or repair.

Note: The user should also examine the shock sensors attached to the shipping carton and notify Verasonics immediately if any of them have tripped, or if there are any other signs of damage. Once the system is unpacked, the user should inspect it for any evidence of physical damage.

1.7 System Transportation

Ensure that all cables are disconnected before packaging the Vantage System.

Always use the original shipping box and the packaging materials provided by Verasonics when transporting the Vantage System. The shipping box is specially designed to protect the system when transporting and has handles for easy lifting.



Foam endcaps are fitted at the front and rear ends of the Vantage System

Figure 1: Verasonics Shipping Box

Note: The user is liable for any damage caused to the Vantage System due to improper packaging. The user should be cautious when lifting the Vantage System—it weighs up to 44 lbs. (20 Kg). A system with the HIFU option installed can weigh up to 57 lbs. (26 Kg). The user should bend their knees and lift the Vantage hardware system from the side with their arms at the front and back, and their fingers extending under the bottom of the chassis.

2. Vantage System Overview

The Vantage System comprises the Vantage hardware system, the Vantage software, and the host controller. The **Vantage hardware system** consists of a single chassis with one or more acquisition boards, and other supporting circuit boards—such as the system Backplane (BKP), Transmit Power Controller (TPC), and a Universal Transducer Adapter (UTA) system or a Scan Head Interface (SHI) module.

The Vantage software consists of the following:

- Verasonics MATLAB® scripts and user-defined scripts, which are the application-level software running on MATLAB.
- Verasonics Hardware Abstraction Layer (HAL)
- Diagnostic tools
- Hardware device drivers
- Code for FPGA devices in the Vantage hardware system—this code must be updated on the Vantage hardware system whenever a newer version of the software is installed.

The **host controller** is a computer on which MATLAB and Vantage software are installed.

Note: The host controller should be used solely for running the Vantage hardware system and should not have other unrelated applications installed on it.

The system requires a user-licensed version of MATLAB and the Signal Processing Toolbox. The user should activate a MATLAB license prior to running the Vantage software.

Note that the monitor is NOT supplied by Verasonics.

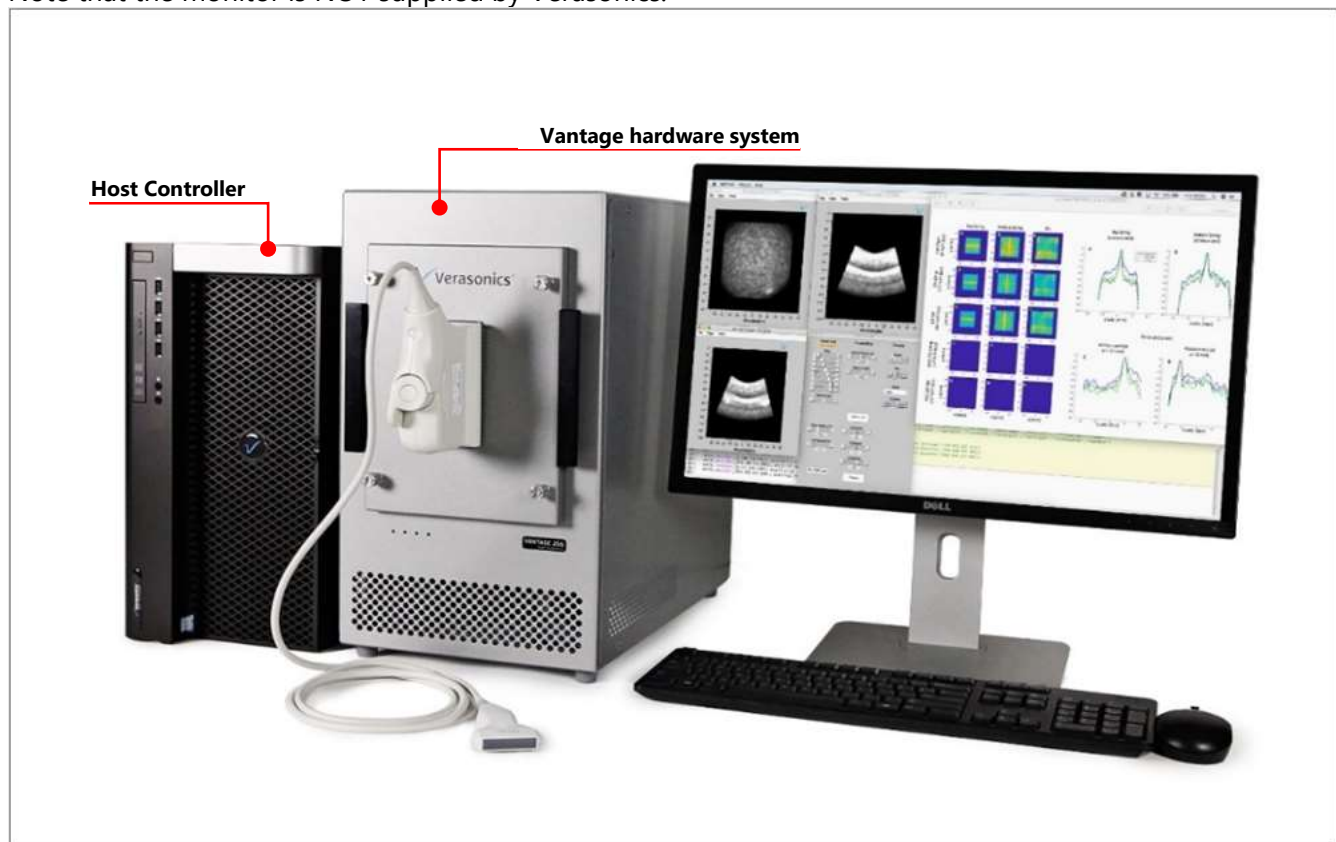


Figure 2: Vantage System

2.1 Vantage Hardware System External Connectivity

This section outlines various interfaces that are available on the Vantage hardware system to facilitate external connectivity to help users implement their own external circuitry.

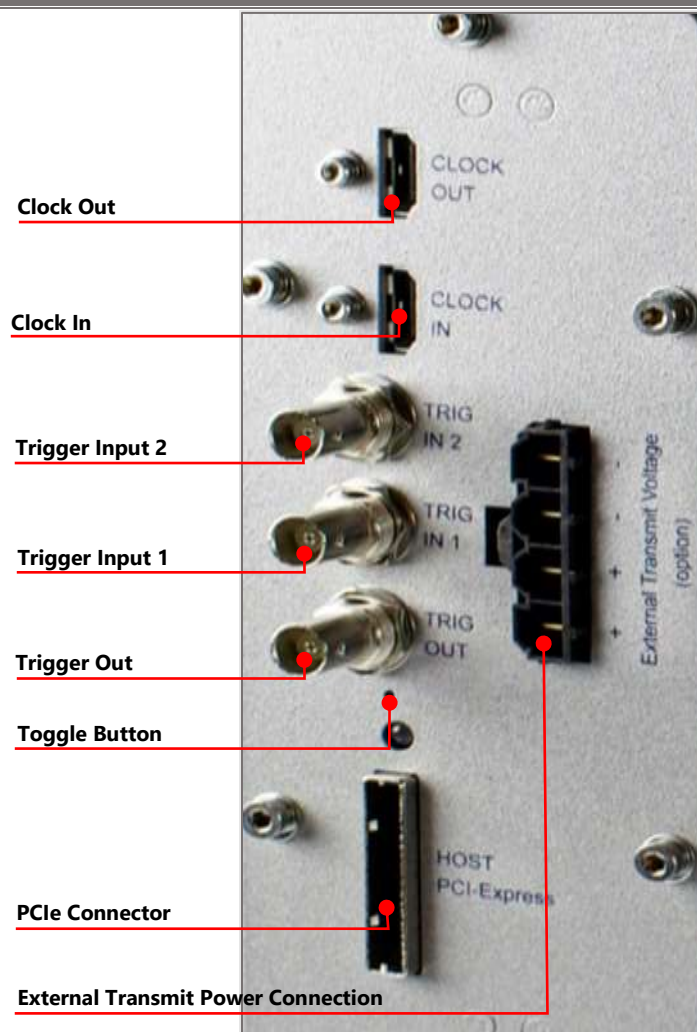


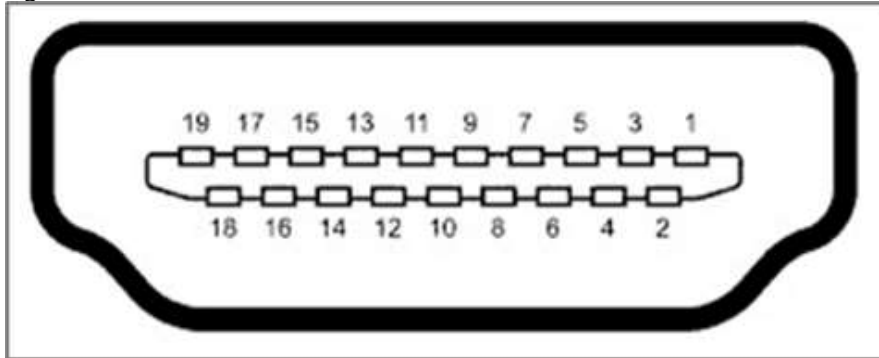
Figure 3: Vantage Hardware System Rear Panel I/O Interfaces

2.1.1 Clock Out

The external Clock Out interface uses a HDMI form factor connector (located at the rear of the Vantage hardware system). This port provides signals that enable synchronous operation with external equipment. This port includes a 250MHz differential CML clock source, a variable differential CML clock source based on the acquisition sampling frequency, one LVTTTL trigger out, and two LVTTTL trigger input signals.

Note: Do not connect the Clock Out interface to a standard HDMI device, as it uses a custom pinout and use model.

Figure 4: Clock Out Pinout



A brief description of the Clock Out pinout is given below.

Table 3: Clock out connector pinout description

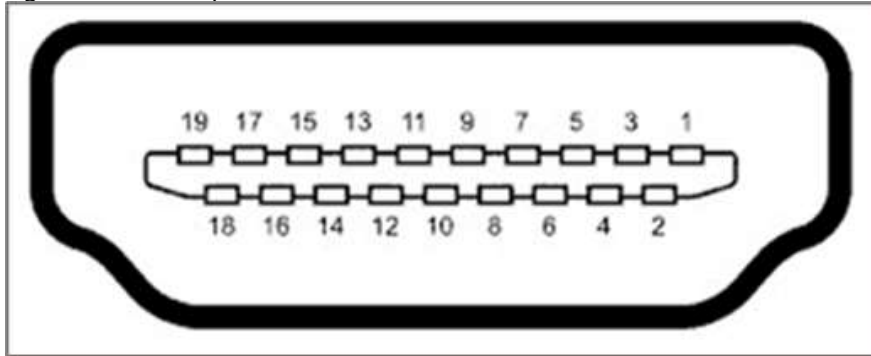
Pin Number	Name	Function
1	250M_OUTp	AC-coupled CML differential pairs for the 250MHz CML clock output
2	Ground	
3	250M_OUTn	AC-coupled CML differential pairs for the 250MHz CML clock output
4	SYNC_CLK_OUTp	Positive member of a differential pair for the Sync Clock CML clock output
5	Ground	
6	SYNC_CLK_OUTn	Negative member of a differential pair for the Sync Clock CML clock output
7	Reserved	
8	Ground	
9	Reserved	
10	Reserved	
11	Ground	
12	Remote Ground	Used for cable detection within the system
13	EXT_FAULTn	Multi-chassis fault signal. A temporary connection to the ground on this shared signal will cause the system to stop imaging. Reserved for use by Verasonics
14	Reserved	
15	Reserved	
16	Reserved	
17	Ground	
18	DIST_PWR *	* Reserved for use by Verasonics.
19	Ground	

2.1.2 Clock In

The external Clock In interface uses the HDMI form factor connector (located on the rear of the Vantage hardware system). This port provides the system capability to synchronize with an external set of clocks and triggers. These clocks are typically provided by another Vantage hardware system but can also be sourced by other external systems.

Note: Do not connect the Clock In interface to a standard HDMI device, as it uses a custom pinout and use model.

Figure 5: Clock Input - Pinout



A brief description of the Clock In pinout is given below.

Table 4: Clock In connector pinout description

Pin Number	Name	Function
1	250M_INp	AC-coupled CML differential pairs for the 250MHz CML clock input
2	Ground	
3	250M_INn	AC-coupled CML differential pairs for the 250MHz CML clock input
4	SYNC_CLK_INp	Positive member of a differential pair for the Sync Clock CML clock input
5	Ground	
6	SYNC_CLK_INn	Negative member of a differential pair for the Sync Clock CML clock input
7	Reserved	
8	Ground	
9	Remote Ground	Used for cable detection within the system
10	Reserved	
11	Ground	
12	Ground	
13	EXT_FAULTn	Multi-chassis fault signal. A temporary connection to the ground on this shared signal will cause the system to stop imaging. Reserved for use by Verasonics
14	Ground	
15	Reserved	
16	Reserved	
17	Ground	
18	DIST_PWR *	* Reserved for use by Verasonics
19	Ground	

2.1.3 Trigger Out

The external Trigger Out signal uses a BNC connector (located at the rear of the Vantage hardware system). This signal is essentially a copy of the LVTTTL trigger output available on the HDMI clock out port. The signal connection is provided in a user-friendly BNC format for instances where the user wants to obtain a trigger from the system.

The output signaling level is 3.3V LVTTTL. This trigger can be generated by the user's imaging script and is typically a 1uS wide, active-low pulse. The trigger 'position', relative to the event in the imaging script, is programmable in several ways. It can contain an absolute delay, specified in 250MHz clock intervals from the time the trigger command was encountered or it can be defined relative to the next acquisition start or relative to an external SYNC_CLOCK rising or falling edge.

2.1.4 Trigger Input1 and Trigger Input2

Both external Trigger Input1 and Trigger Input2 use BNC connectors (located at the rear of the Vantage hardware system). This signal is essentially a copy of the LVTTTL trigger inputs available on the HDMI Clock Out port. The signal connection is provided in a user-friendly BNC format for instances where the user wants to obtain a trigger from the system.

The input signaling level is 3.3V LVTTTL but can handle positive going signals up to 5V. The trigger condition is configurable with either rising or falling edge detection. If both inputs are defined, they will need to be phase aligned for the chosen edge on both inputs. The trigger can be chosen to be required for the sequence to continue or it can be configured so that the sequencer continues up to the point of a RUN_ACQ instruction and is only required to begin at the beginning of that event. These trigger inputs are sampled based on the rising edge of SYNC_CLK_IN. If SYNC_CLK_IN is not active, the hardware automatically switches to sampling relative to the SYNC_CLK_OUT. A timeout of up to 60 seconds can be specified in 250 msec increments. This timeout allows the sequence to continue if the external trigger is not connected and can be configured to either report the timeout condition or not.

The SYNC_CLK_OUT rate is given by 1/4 the A/D converter sample rate; therefore, it is dependent on the user control script settings. The following relations apply:

$VDASADClk = 250 / (VDASADRate)$, where 250 is the system clock frequency in MHz.

$SYNC_CLK_OUT = VDASADClk / 4 \text{ MHz}$

For more details, refer to the [Sequence Programming Manual](#).

2.1.5 PCIe Connection

The system uses a PCI-express interface for connection (through an extension cable and adapter board) to an expansion slot in the host controller. All interaction with the host controller passes through this interface, including the DMA transfer of receive data from the Vantage hardware system to the host controller, and programming data, commands, and parameter values from the computer to the system.

The cable must be plugged into the Verasonics Isolating Host Adapter module installed in the host controller. The PCIe expansion slot used for the Vantage Host Adapter card must provide either **16 lanes at PCIe Gen 2 speed, or 8 lanes at PCIe Gen 3 speed**. The system will be fully functional in an expansion slot with reduced lane width or speed but at a reduced DMA transfer rate performance.

A 1-meter PCIe interconnect cable is provided with the Vantage System, for the connection from the host adapter card to the Vantage hardware system. It is possible to use a longer interconnect cable, but some customization of system settings may be required for reliable operation at full bandwidth for a different cable. Contact Verasonics for additional details.

2.2 Front Panel Status Indicators

There are four LED indicators visible on the front side of the Vantage System chassis that can provide information on the operating state of the Vantage hardware system.

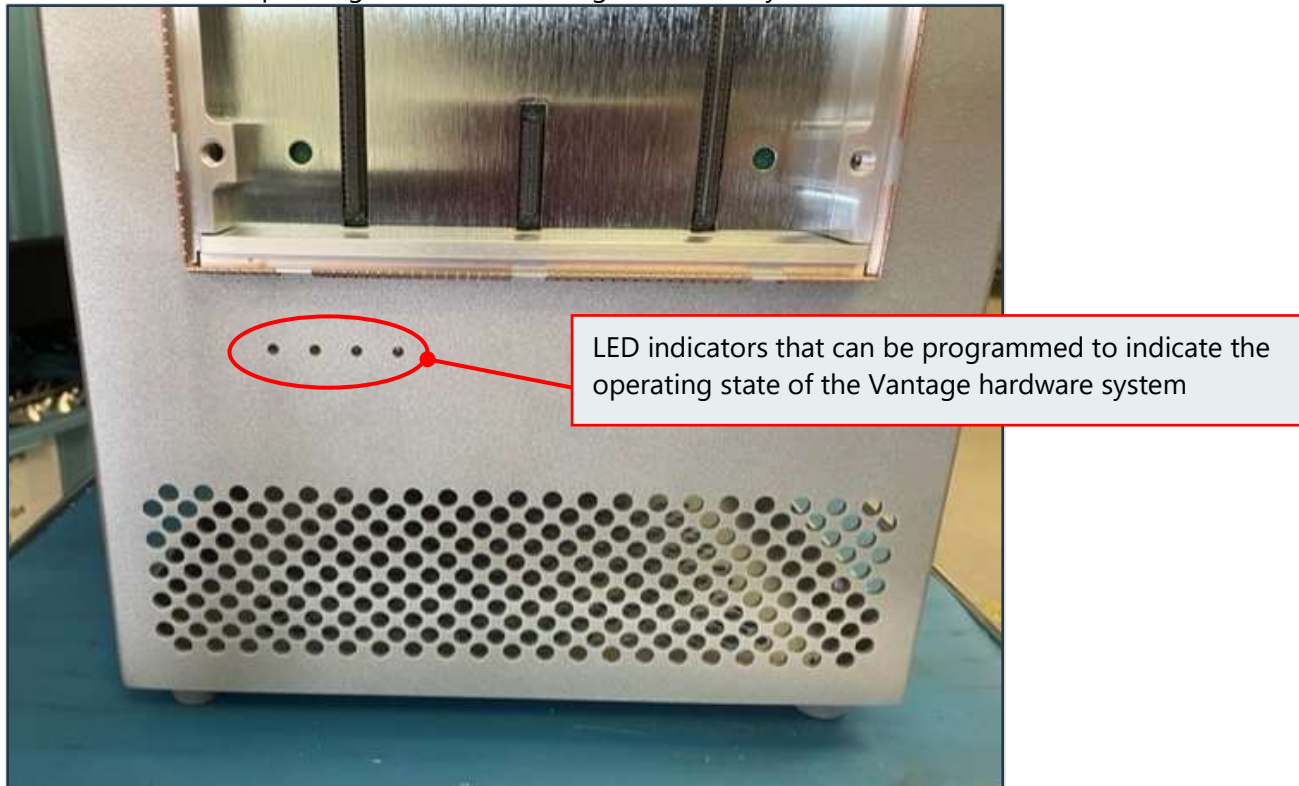


Figure 6: Front Panel Indicators on the Vantage hardware system

Several status signals are available to drive these LED indicators. The user select any desired signal to map to each of the four LEDs from within the Setup script by specifying the field,

`"Resource.Parameters.SystemLED"`.

If the `"Resource.Parameters.SystemLED"` field is not specified in a SetUp script the system automatically creates it with default values of `{'running', 'paused', 'activeTxAndOrRx', 'transferToHostComplete'}`.

To use this feature, the variable `"SystemLED"` must be defined as a 1 X 4 cell array of strings, where each of the four strings specify the signal to be assigned to the associated front panel LED indicator from left to right. The string values that can be used are listed below:

activeTxAndOrRx - LED is ON while the hardware sequencer is executing a transmit-receive event.

activeTxAndOrRxProfile5 - LED is ON while the hardware sequencer is executing a transmit-receive event that uses TPC profile 5 (Extended Transmit or HIFU).

externalBoxFault - Fault condition in external sync box.

missedTimeToNextAcq - LED turns ON when a specified time interval is not achieved.

missedTimeToNextEB - LED turns ON when the specified time interval is not achieved.

off - LED is turned OFF.

paused - sequence has paused for any of the individual reasons listed below:

- **pausedDmaWaitPrevious** – sequence has paused waiting for the previous DMA to complete.
- **pausedDmaWaitForProcessing** – sequence has paused waiting for DMA processing complete signal from the software event sequence.
- **pausedOnMultiSysSync** - pause waiting for trigger through the sync box.
- **pausedOnTriggerIn** - pause waiting for the trigger.
- **pausedSync** - pause waiting at a Sync command.

running - LED is ON while a hardware event sequence is running.

starting - LED turns ON momentarily while the hardware event sequence is starting to run (for example, when coming out of the "freeze" state).

transferToHostComplete - a DMA transfer of RF data from the Vantage hardware system to the host controller is underway or has recently been completed.

Note that some of the hardware states listed above have very short durations (a few milliseconds or less) that would make the ON state of the associated LED imperceptible. In these cases, the system automatically increases the ON-state duration for the LED to several tenths of a second, so that the user can see it.

2.3 Probe Connector Illumination

Systems configured with the Cannon DL-260 or DL-360 probe connectors provide an LED located behind the hole in the connector for the ZIF latching mechanism. This LED is intended to make it easier to locate the connector when using the system in a darkened room. By default, the system enables this LED at all connectors, when nothing is connected to them.

A new field "Resource.Parameters.ProbeConnectorLED" can be used to disable the connector LED. This field must be set to a 1X4 array of MATLAB doubles, with each value to set to 1 to enable the associated connector's LED or 0 to disable it. The four values represent the four possible connectors supported by the software; on a single-connector system the last 3 values are ignored or on a 2-connector system the last 2 values are ignored.

If you do not specify this field in your script, VSX creates it and assigns a default value of [1 1 1 1] to enable all connector illumination LEDs. To disable the LEDs, simply specify as

`Resource.Parameters.ProbeConnectorLED = [0 0 0 0]` in the SetUp script (or any other combination of 1's and 0's as desired).

Note: This feature only applies to the Cannon DL-260 or DL-360 probe connector types. The UTA modules—UTA 408, UTA 408-GE, and UTA 160-DH/32 do not provide connector illumination LEDs at all so the `ProbeConnectorLED` field has no effect for those modules.

3. Documentation

3.1 Documentation Included in the Vantage Software Release Folder

This section provides a list of documents located in the *Vantage project/Documentation* folder. All documentation is in PDF format; it is not required to install or run MATLAB to access these files. The documentation folder is accessible to users after Vantage software installation.

Vantage System User Manual

The Vantage System User Manual provides important safety and regulatory guidelines for proper use of the system; an overview of the external connectivity options and front panel status indicators; a synopsis of the transmit and receive path functionality and system features; the summary of available documentation; detailed instructions on software installation, final setup, and tests to verify system functionality; information on backward compatibility constraints associated with certain software releases and system hardware configurations; important information on licensable software capabilities and software license management. The Vantage System User Manual provides essential information and instructions necessary for the proper understanding and operation of the system. It is highly recommended that the user carefully reads the user manual.

Vantage Product Specification

This document provides a detailed description of the features, options, and performance specifications of the Vantage System.

Vantage System Initial Setup Instructions

This document provides a quick summary on unpacking, connecting, and starting up the Vantage System. A copy of this document is also included in the shipping carton and is shipped alongside the Vantage System.

Vantage System Troubleshooting Manual

This document identifies common operator errors and provides tips and techniques for avoiding these errors, correcting problems with the Vantage hardware system, and identifying and correcting programming errors in user-defined SetUp scripts.

Vantage Sequence Programming Manual

This document is a comprehensive reference guide to programming the Verasonics System from within the MATLAB scripting environment.

Vantage Tool Summary

This document provides an overview of each of the tools available in the *Tools* subfolder within the Vantage *Project* folder.

Vantage System Release Notes

This document covers important release information describing new features, bug fixes, and known errata with the current software release.

Vantage System Risk Analysis

This document is in the *Safety* subfolder (File path: *Vantage project/Documentation folder/Safety*). This document outlines the risk analysis that has been conducted by Verasonics for the Vantage System.

Application Note: Vantage Example Scripts for NDE/NDT Applications

This document provides a matrix summary of all available example scripts in the directory: Examples_NDE_NDT.

Application Note: Vantage Example Scripts for Biomedical Applications

The document provides a list of example scripts associated with biomedical applications.

Verasonics Third-Party Software Notices

This document outlines third-party terms and conditions applicable to certain code, software, and technologies, which may be used in one or more Verasonics software products.

3.2 Other Documentation Included in the Software Release Package

Additional documentation can be found at other locations within the *Vantage Project* folder. You can locate the documentation for software utilities within the folder where the software utility is saved.

Script Matrix and Description This document can be found in the *Example_Scripts* folder within the Vantage Project folder. It provides a cross-reference summary of the types of example scripts included in the *Example_Scripts* folder, and which of those types are included in the folders for each of the individual Probes supported by the system. A one-paragraph summary description of each script type is included in the document as a matrix. By reviewing this matrix, you can easily identify and locate scripts you may be interested in without having to browse through the individual probe directories. In addition, a more detailed description of many example scripts can be found in the comments included in the example script itself.

Specialized Applications Example scripts for several specialized applications, and/or associated features of the Vantage System, can be found in the *Example_Scripts/Specialty_Applications* subfolder of your Vantage Project folder. Many of these example subfolders include additional pdf documentation or application notes files.

3.2.1 System Software Tools and Utilities

A number of useful tools and utilities can be found in the Utilities and Tools subfolders within the Vantage project folder.

The *Tools Summary* document (located in the *Vantage Project /Documentation* subfolder) provides an overview of the available tools.

For the Utilities folder and additional high-level system management and self-test utilities located at the top level of the Vantage project folder. This section provides a brief description of software utilities supported by the Vantage System.

- **Version**

This function can be invoked from the MATLAB Command prompt. It provides a quick summary of the system software and hardware configuration and software license status—these are all displayed in the MATLAB Command Window. It is a valuable check to use as quick confirmation that there are no obvious problems with the overall state of the system. A more comprehensive system configuration report can be obtained by adding an argument to the function call, for example by typing `Version x` (the value or character used as the argument does not matter). In this case, the output report will include the part number and serial number of all major hardware modules in the system, FPGA code revision levels, etc. Version will also check the status of the FPGA code installed in the Vantage hardware unit and ask the user whether to proceed with updating any FPGA revisions found that are incompatible with the SW release being used.

- **reprogramHardware**

This utility will force revision checks and updates to all FPGA code files in the HW system. It can be used to recover from a situation where one of the installed FPGA files has been corrupted, preventing high-level functions such as Version from functioning properly (for example, if FPGA updates through the Version command did not complete successfully because the system was shut down prematurely while the updates were still in process). When using the `reprogramHardware` utility, pay attention to the prompts and status updates displayed in the MATLAB Command Window. In some cases, an FPGA file update will take several minutes, and the system must not be interrupted while the update process is underway. After all updates are completed, you will be prompted to shut down the system to a full power-off state. A full power cycle is required for the newly loaded FPGA files to take effect.

- **ProbeConnectorStatus**

Typing this command at the MATLAB command prompt results in a display of the status of the probe connector(s) on the system, including the connector type, whether a probe is connected, and if so the ID and name of the transducer (if the ID is recognized by the system).

- **VerasonicsVerificationTest** or **VVTA** comprehensive self-test of the entire system. See the "Verasonics Verification-test" heading within the "Final Setup and System Tests" section of this document, for instructions on running `VerasonicsVerificationTest`.

- **SystemInfo**

This utility provides a comprehensive status report of the configuration of your host computer, Vantage SW installation and SW license status, and Vantage hardware system (if one is connected). Refer to the "Software Licensing" section of this document for more information on SW licensing and how to use the SystemInfo utility.

- `SwitchesInfo`

This utility generates a report of the PCIe switch devices and associated links in the PCIe communication path between the host computer and the PCIe device endpoints within the Vantage System. You can use this utility to confirm all PCIe links are operating properly at the intended lane width and speed grade, and whether an excessive error rate is occurring on any of the links.

- `ExternalClockSelect`

This utility allows the user to switch between the system's internal 250 MHz system clock source or an external clock source connected to the "Clock In" connector on the rear panel, without having to exit MATLAB or cycle system power. Note that this utility can only be used from the MATLAB command prompt, while no other script is running. Simply type `ExternalClockSelect` at the MATLAB command prompt, hit return, and it will report the clock source currently being used, and then query whether you want to switch or not. If you ask it to switch to an external clock source it will first check for the existence of a valid 250 MHz signal from the connector, and then either inform you that an external clock signal is not present or make the switch and confirm the result.

- `TransmitWaveformUtils`

This subfolder includes utility functions you can use to manipulate transmit waveforms for use with the Vantage System, and for converting waveforms from one representation to another. Starting with the 4.0.0 release, `States` is the array format used by the Vantage software to represent a transmit waveform. In earlier releases, a `PulseCode` format was used. However, the `PulseCode` format is still used by some Vantage software utilities and examples and is fully supported by the system.

The `TransmitWaveformUtils` subfolder includes conversion utilities to convert a `PulseCode` array into the `States` format, and also to convert back and forth between the `States` array format and a "TriLvl" direct vector representation of the waveform, useful for displaying the waveform graphically and for analyzing or synthesizing waveforms using MATLAB signal processing utilities. Refer to the comments at the beginning of each of these conversion utilities for definitions of the input and output arguments, and to the Sequence Programming Manual for more information and definitions of the `States` waveform representation.

3.3 Documentation Available on the Verasonics Community Portal

This section provides a list of documents available on the Verasonics Community Portal.

Application Note: Wall Filter Frequency Responses Under CDI Weak/Low Flow Conditions

This document describes the performance of weak/low flow wall filter coefficient sets available to the Doppler Process Object defined in the Sequence Programming Manual.

Application Note: Volume Imaging Package (4.5.3)

The document introduces the user to the Multi-System Volume Imaging Vantage product and provides details on how to properly write a script to program and operate the multi-system configuration. The document provides an overview, hardware setup, sequence programming details, remote commands, initiating a sequence, and a list of known bugs and issues.

Application Note: Vantage Analog Signal Path Gain

This document provides a summary of the analog signal path gain of the Vantage System.

Application Note: Use of HVMUX Transducer with Vantage System

This document describes user-programmable features included in the Vantage System that enable power and control for HVMUX switches located within a transducer.

Application Note: New UI Scheme 4.4

This document outlines a modern object-oriented design for defining UI controls. The new design brings together the flexibility and familiarity of defining sequences, with new features that make script development easier.

Application Note: ShowTXPD Tool

The showTXPD tool enables users to visualize the alignment of the transmit beam with the pixel data. This document describes the features of the showTXPD tool.

Application Note: Shear Wave Visualization Script - Description and Operation

This document describes how to configure and use the shear wave visualization script. Important considerations prior to the use of this script are also provided.

Application Note: Quadrature Sampling

This document provides analysis and illustrations to show the generation of true analytic signals with direct quadrature sampling.

Application Note: Process Storage Class - Data Import and Export

This document outlines the functionality of the process storage class, which provides an efficient method for defining sequence-programming-based RF and image-data file import and export. In addition, it describes parameters that are supported by the process storage class.

Application Note: Probe Temperature Sensing

The Vantage System provides hardware circuitry and software utilities to allow monitoring of one or two transducer temperature sense thermistors connected through dedicated pins at the probe connector, for probes that provide this feature. The system can be programmed to automatically monitor the temperature sensors while a script is running and exit the script with an error condition if a predetermined temperature limit is exceeded. The temperatures can also be read from the Vantage hardware system whenever desired, for display or other purposes. At the hardware level, the system measures the resistance of the device connected to the thermistor signals at the probe connector. If that device is a thermistor, the user can translate the measured resistance into a temperature but can also use this feature to measure the resistance of some other type of sensor or control within the probe. This document describes the functionality of the hardware feature, how to program it for monitoring and for enabling the automatic shutdown threshold, and how to convert the monitored value into a temperature.

Application Note: Phase-Accurate Synchronization of Multiple Vantage Systems Using the Multi-System Synchronization Module

This document describes the use of the Synchronization Module hardware and software configuration needed to share a single 250 MHz system clock between multiple Vantage Systems, and thus enable synchronous and phase-aligned operation of multiple hardware sequencers. This arrangement permits coherent acquisitions with up to 2048 channels (using up to 8 Vantage 256 systems; it is also possible to synchronize different Vantage models). Each Vantage System operates with its own host controller when running by itself, and the data collected by each system can be processed as is normally done for real-time display. The data can also be transferred to a common host controller during the sequence or after the experiment and processed coherently using the complete data set.

Application Note: NDE Research Software

The document introduces users to the NDE Research Software and provides instructions for its use. The features of the software are described, and each user setting is explained. In addition, instructions and examples are also provided for the development of custom imaging modes.

Application Note: Multithreading Framework

The document outlines the implementation of a multithreading framework for parallelizing code and fully leveraging the benefits offered by multicore and multi-CPU systems.

Application Note: Introduction to Waveform Generation Applications in Ultrasound

This document introduces waveform generation applications in ultrasound and provides an overview of conventional transmit waveforms and coded excitations.

Application Note: Vantage HIFU Configuration Setup Instructions

This document provides instructions on the wiring connections, initial setup, and functionality testing of the AIM-TTI QPX600DP external transmit power supply for use with a Vantage System configured with the HIFU transmit option.

Application Note: Extended Input/Output

The document introduces users to the Extended I/O option functionality and typical use cases. Detailed instructions are provided for the installation and programming of the option, including the use of setup objects and process functions. To aid understanding, this document is best used in conjunction with the provided example scripts.

Application Note: Event Analysis Tool

This document provides an overview of the Event Analysis Tool that can assist with developing and debugging a setup script. In addition, it offers debugging strategies for identifying errors in the script and describes the capabilities of the tool that can be used for debugging setup scripts.

Application Note: DC Bias for Transducer Element

This document summarizes how to apply DC bias voltage to transducer elements while using a Vantage hardware system.

Application Note: Cannon Connector to Cable Interface PCB - Signal Breakout Board

This document specifies the requirements and design for an interface PCB assembly and associated shields to allow convenient connection of cables to the transducer connector used on the Vantage hardware system.

Application Note: Arbitrary Waveform Generation Toolbox

This document describes the use of the Verasonics Arbitrary Waveform Generation Toolbox, called the GUI-based arbitrary waveform generation tool (GUI-AWE), version v1.

Arbitrary Waveforms using a Tri-State Transmit Pulser

The paper outlines methods for encoding waveforms into sequences suitable for a tri-state RF ultrasonic transmitter, under various fidelity criteria.

Application Note: Use Models and Fidelity Criteria for the Verasonics Arbitrary Waveform Encoders

This document explains the use models and fidelity metric concepts needed for users of the Verasonics Arbitrary Waveform (AW) Encoder tools.

Application Note: UTA 1024-MUX Adapter

This document describes the intended use of the Vantage UTA 1024-MUX adapter, the required configuration, programming model, assembly, and the basic sequence needed to complete a 1024-element acquisition made up of multiple transmit-receive events.

Tutorial for Classifying Objects in Ultrasound Data

This tutorial is designed to be used as a guide on how to use the MATLAB® Statistics and Machine Learning toolbox in combination with the Verasonics Vantage System for live classification. It contains four easy steps that allow users to understand the overall classification workflow. The parameters in this tutorial work best for classifying the large hyperechoic circles in images from a CIRS Multi-Purpose Multi-Tissue Ultrasound Phantom Model 040GSE. This tutorial requires the use of an L11-5v transducer, MATLAB's Statistics and Machine Learning Toolbox, and MATLAB's Image Processing Toolbox.

FUS Elite User Manual

The FUS Elite User Manual contains important safety and regulatory guidelines for proper use of the system, component descriptions and specifications, instructions on how to use the graphical user interface to perform Ultrasound Guided Focused Ultrasound (USgFUS) therapy planning and delivery, guidelines on experimental data collection and data management capability.

408-GE Backshell Kit Assembly Instructions

This document provides assembly instructions for the backshell kit and guidelines for wiring element signals to pads on each of the cable termination boards.

Glossary of Special Terms Used by Verasonics

This document contains a list and explanation of special terms used by Verasonics.

Probe Impedance Check User Guide

This document outlines the Probe Impedance Check tool, which is intended to provide users with a quick and simple way to measure the impedance of each element of a probe that is connected to the system. It is located in the *Tools* folder, at *Tools/SimpleImpedance_Tool*. It identifies any suspected shorts to ground (measured impedance less than about 10 Ohms) or open circuits (measured impedance greater than several hundred Ohms).

Vantage System 260-Pin ZIF Transducer Connector

This document provides information regarding the usage model of the ITT-Cannon 260-pinZIF transducer interface connector used on the Vantage System.

4. Software Installation Requirements

When a new Vantage software version is released and becomes available in the *Customer Download* folder, it is highly recommended to review all documentation (especially, the [Vantage System Release Notes](#) associated with the release) before installing the software.

The current Vantage Software release has been tested for compatibility and full functionality with the versions of MATLAB and host controller Operating Systems listed in Table 5.

Table 5: MATLAB and host controller OS requirements

Software	Version
MATLAB Version	Allowed: R2020b - R2022a Recommended: R2022a
Device Driver	Windows WinDriver: 12.9.0 Linux WinDriver: 15.0.0
Host Controller OS Version	
Windows OS	Windows 10 2016 LTSC
macOS	10.10 (Yosemite) and greater simulation only
Linux OS	18.04 LTS w/ Kernel version 5.4 (the default)

The software installation and setup procedures vary based on the following scenarios:

- Using a Verasonics-supplied host controller with a new Vantage hardware system
- Using a customer-supplied host controller with a new Vantage hardware system
- Updating a host controller with the latest software release version
- Using a simulation-only software license on a laptop or desktop computer

The following sections provide more detail on software installation requirements and setup procedures associated with each of the above scenarios.

4.1 Using a Verasonics-Supplied Host Controller with a New Vantage Hardware System

The latest Vantage software version is installed on a Verasonics-supplied host controller before it is shipped to the customer; therefore, Vantage software installation is not required. To set up the system, follow the instructions in Section 5.

4.2 Using a Customer-Supplied Host Controller with a New Vantage Hardware System

Verasonics supports the use of a Vantage hardware system with a customer-supplied host controller. This requires configuring the host controller for use with the Vantage hardware system prior to software installation. Refer to the [Vantage Release Notes](#) document for information on system requirements a host controller must meet for use with the current Vantage software release version.

Note: It is important to verify that the versions of OS and MATLAB installed on the host controller are compatible with the current Vantage software release. Refer to [Vantage System Release Notes](#) for recommendations on versions of the MATLAB and host controller OS that are compatible with the current Vantage software release.

In general, Vantage software, which runs on the host controller from within MATLAB, benefits from faster CPU speeds, a greater number of CPU cores, additional RAM, and faster graphics displays. The Vantage software is highly parallelized and optimized to use as many CPU cores as are available.

After the initial system setup of the Vantage System, install Vantage software on a customer-supplied host controller. For software installation instructions, refer to Section 5.

4.3 Updating a Host Controller with a New Vantage Software Release Version

On a fully configured Vantage System running an earlier release of the Vantage software, install the latest software release. The software installations from any earlier Vantage software releases can be left in place so that the user can easily switch back and forth between software installations. To install Vantage software, follow the instructions in Section 5.

Note: Different software releases can require a different version of the driver for communication with the Vantage hardware system; however, the driver lacks the capability to accommodate the installation of multiple driver versions concurrently. Therefore, Verasonics does not support the installation of multiple versions of the driver.

4.4 Installing Simulation-Only Software on a Laptop or Computer

The customers who have purchased simulation-only software licenses from Verasonics for use on a laptop or a customer-supplied host controller (that is not attached to the Vantage hardware system) need not install the driver after the software installation process. The absence of a driver indicates to the Vantage software that this is a simulation-only installation that does not interface with the Vantage hardware system.

To install Vantage software, follow the instructions in Section 5.

Note: Before performing the Vantage software simulation-only installation for the first time, the user should obtain an updated license file from Verasonics—this license is specific to the personal computer or laptop on which Vantage simulation-only software is installed.

Note, however, that the user can complete the software installation, and run utilities such as Version and systemInfo, even if the computer/laptop does not yet have a valid simulation license. After the software installation is complete, the user can run the systemInfo utility to obtain the information needed to request an updated license file from Verasonics.

For more details on the Vantage software simulation-only license, refer to Section 8.3.

5. Software Installation Instructions

Follow the software installation instructions provided in this section to install Vantage software on a customer-supplied host controller or to install a newer version of Vantage software on a host controller. Note that it is not required to install software on a new Verasonics-supplied host controller as the latest version of the software is preinstalled on it.

5.1 Download the Vantage Software Distribution and License

The latest Vantage software distributions are available in the *Vantage Releases* folder within the Verasonics *Customer Download* folder. If the user requests a copy of an older release, it should be stored at a different location within the *Customer Download* folder. Periodically, Verasonics updates the *Vantage Software Release* folder with new Vantage software distributions.

To download a software distribution:

1. Navigate to <https://downloads.verasonics.com>, and enter the username and password.
If you have trouble accessing your *Customer Download* folder, please contact the Verasonics Technical Support team (support@verasonics.com) for assistance.

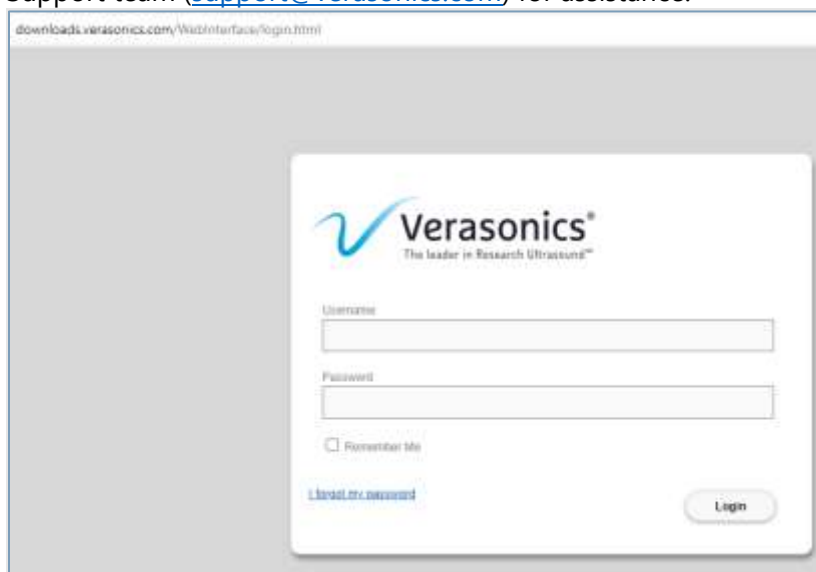


Figure 7: Verasonics downloads webpage

2. After logging in, locate a folder at the top level, clearly labeled with the customer name. Within that folder, there is a license file and *Vantage Releases* folder.

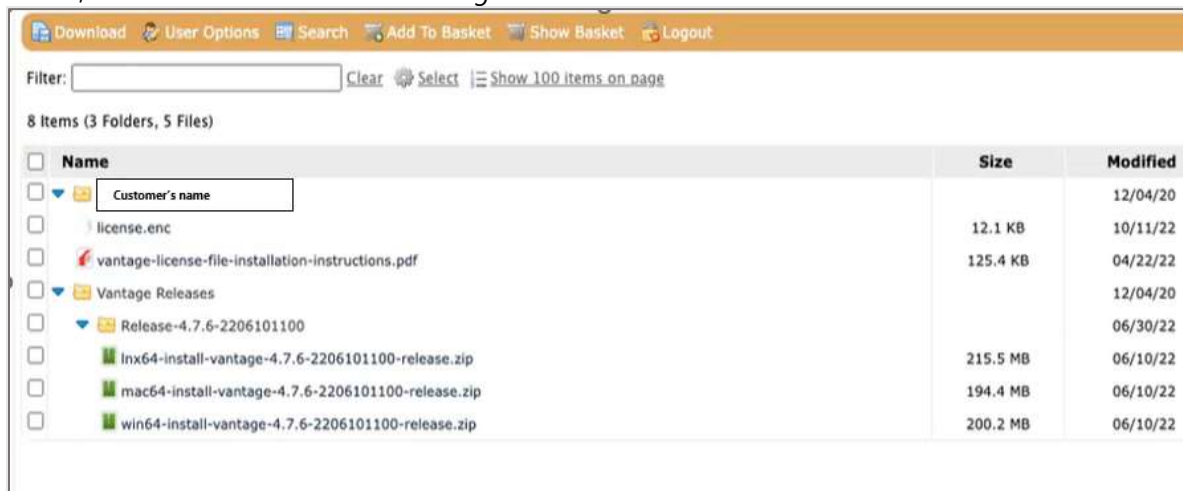


Figure 8: Customer folder located on the Verasonics download webpage

3. Within the *Vantage Releases* folder, there is a *Release* folder that contains the latest Vantage software distribution. Note that the Release folder (e.g., *Release-4.7.6.2210101100*) contains three zip archives.

Each of the zip archives is intended for a specific OS platform.

Platform	Vantage Software Distribution Filename
Windows	win64-install-vantage-4.7.0-2112230915-release.zip
macOS	mac64-install-vantage-4.7.0-2112230915 -release.zip
Linux	lnx64-install-vantage-4.7.0-2112230915 -release.zip

Table 6: Release Folder - Zip Archive Options

- Download the software distribution that is suitable for the host controller OS and the current license.
- Copy the software distribution file (Windows, macOS or Linux) to the root of the *Vantage project* folder.

5.2 Install the Vantage Software Release

Installing a new Vantage software release does not impact any other Vantage software installations that may be present on the host controller. The software installation process does not interact with the Vantage hardware system, so the user can install software on a host controller with or without connecting it to the Vantage hardware system.

Note: Before installing the software, the user should ensure that there is a backup of the host controller.

To install the software:

- Exit any MATLAB applications or any other non-essential applications that may be running on the host controller.
- Copy the downloaded software distribution file to the desired installation location.
- Vantage software distributions are Zip archives. Double-click on the distribution file to open it.
- Extract the distribution.
- Copy the *license.enc* file to the root of the Vantage project folder. Starting with release 4.3.1, Vantage software distributions do not include a customer-specific license file.

Note: It is not required to install a driver if Vantage software is used for simulation-only purposes (i.e., if the host controller is not connected to a Vantage hardware system). For instructions on how to configure MATLAB to use Vantage software, proceed to Section.

- Install the Windriver device driver. For instructions on driver installations, refer to Section 5.4.

5.3 About Vantage Project Folders

The main directory containing all Vantage software installation files (*Vantage-4.7.6-2206101100*) is referred to as *Vantage project* folder across all documentation.

- Vantage project folders are self-contained entities. Vantage project folders are created either by unzipping a Vantage software distribution or by copying an existing *Vantage project* folder. The first *Vantage project* folder is created on a host controller by unzipping a Vantage software distribution unless the *Vantage project* folder is copied from another computer system.
- The user can create any number of *Vantage project* folders. They can be located anywhere on the host controller system. The user can also rename them to identify different installations for each individual user of the system or to identify several distinct projects. For instructions on configuring and managing multiple software installations on the same computer, refer to Section 10.1.
- The *Vantage project* folder is created by unzipping a Vantage software distribution and is named Vantage-[release number].
- The user should never save a *Vantage project* folder on a filesystem that is not local to the host controller, such as a filesystem managed by Microsoft OneDrive or Apple iCloud Drive.
- The Documentation folder is found in the *Vantage project* folder after the Vantage software installation. All documentation is in PDF format; there is no need to install or run MATLAB to access these files.

5.4 Install the WinDriver Device Driver

It is necessary to install a device driver on a customer-supplied host controller to enable communication between the host controller and the Vantage hardware system.

- On Linux or Windows OS-based host controller, install a driver using the instructions provided in Section 5.4.1.
- A macOS-based host controller can only be used for simulation purposes and cannot be connected to the Vantage hardware system; therefore, it is not required to install a device driver on it.

If there is a problem with the driver installation, or if it needs to be updated to a more recent version, follow the instructions provided in this section to overwrite the older driver installation.

Note: The user must complete the Vantage software installation as explained in the previous section before installing the driver.

5.4.1 WinDriver Installation Procedure for Windows-based Systems

The host controller is required to operate on Windows 10 or later versions of the Windows operating system. This release and subsequent distributions do not support older versions of the Windows operating system.

Install the WinDriver device driver on your operating system (Windows 10 or later versions) using the following instructions:

1. Use the Windows File Explorer to navigate to the folder named "*System\wd-install\windows64*" in your *Vantage project* folder.
2. From Windows File Explorer, double-click on the file named "*VSWD_12.9.0.msi*" or right-click on the file and select Install from the context menu. Administrator privileges are required to install the driver.
3. Review the terms of the software license agreement. After agreeing with the terms, click on the check box indicating an agreement and continue with the software installation.
4. Do not change the location where the driver is installed.
5. Answer in the affirmative about installing a device driver from Verasonics, Inc. The installer will open and close a few windows while it does its work. Allow the installer to work to completion.
6. Fully power down and then reboot your host controller before attempting to use your Vantage software.
7. After rebooting the Vantage System, the Vantage hardware should be visible in the Windows Device Manager.

5.4.2 WinDriver Installation Procedure for Linux-based Systems

The officially supported kernel is the 5.4.0 kernel which is the default on Ubuntu 18.04.

Note: For GPUDirect support, the GPU driver must be installed before installing WinDriver.

1. Install the NVIDIA CUDA driver to enable GPUDirect support. GPUDirect is an optional feature that must be licensed by Verasonics. Contact Verasonics support for more information if you are interested in adding this feature.
2. Verify that you have booted to the default kernel for Ubuntu 18.04 (5.4.0). Open a terminal, run the following command, and verify that you get a similar output:

```
$ uname -r
```

```
5.4.0-124-generic
```

The -124 value may differ depending on the patch level of your Ubuntu 18.04 installation. It is recommended that you install all updates and reboot, if necessary, before proceeding to the next step.

```
$ sudo apt update && sudo apt upgrade
```

3. Install the driver.

```
$ sudo System/wd-install/linux64/wd1500Installer.run
```

4. Fully power down and then reboot the host controller before attempting to use the Vantage software.

6. Final Setup and System Tests

After completing the initial setup and software installation (if required) described in the preceding sections, the following tests must be conducted to ensure that the Vantage System setup is complete. There are significant updates in the new Vantage software releases. It is important for users (including existing users who are familiar with earlier Vantage software releases) to follow the instructions provided in this section carefully.

6.1 Activate the MATLAB License

Before using the Vantage software, the user must have the MATLAB application and the Signal Processing Toolbox installed on their host controller. The host controller supplied by Verasonics is pre-installed with MATLAB and the Signal Processing Toolbox and requires activation with the user's license. Refer to the Mathworks© website (<https://www.mathworks.com>) for further details on obtaining and activating a MATLAB license.

6.2 Activate the Vantage Project Folder

The *Vantage project* folder must be activated before it can be used. From within the MATLAB environment, activate your *Vantage project* folder:

1. Set the MATLAB working directory to the root of the *Vantage project* folder.
2. Issue the `activate` command from within the MATLAB Command Window. This function adds the *Vantage project* folder and its subdirectories to the MATLAB path and loads the libraries used by the Vantage software.

The above procedure must always be completed when the user starts MATLAB or switches to a different *Vantage project* folder.

When the user switches between Vantage project folders, they should perform the following steps if the `activate` command was issued in the folder that the user is switching from. This procedure ensures that shared libraries and MATLAB extensions loaded in the MATLAB application memory space are of the same revision level.

To issue activate command from a new folder:

1. Quit the MATLAB application.
2. Restart the MATLAB application.
3. Select a new *Vantage project* folder.
4. Set the MATLAB working directory to the root of the new *Vantage project* folder.
5. Issue the `activate` command in the new folder.

The host controller does not need to be restarted when switching between the Vantage Project folders.

Note: Failure to perform the above procedure can result in erratic operations or the MATLAB operation can crash.

When the user switches between different Vantage software versions that require different FPGA files in the Vantage hardware system, the Vantage software prompts an FPGA firmware update. This update takes effect only after a system power cycle.

6.3 Verify the Vantage Software Functionality

1. After activating a project folder, issue the `Version` command at the MATLAB command prompt.
2. Confirm that the software release version number is reported correctly and that no errors or warning messages are displayed.

If any error or warning messages are reported, contact Verasonics Technical Support (support@verasonics.com) for assistance.

Do not attempt to use the Vantage System if the integrity of the installed software is unclear.

6.4 Update Vantage Hardware FPGA Firmware

Switching to a different version of the Vantage software distribution may require updating the Vantage hardware FPGA firmware. Each version of the Vantage software distribution is compatible with a single version of the Vantage hardware FPGA firmware. Two different Vantage software distributions may use the same version of the FPGA firmware, so an update is not always required. If a version of the FPGA firmware is detected that is incompatible with your Vantage software distribution, a warning message is issued when executing a script or running the `Version` command.

Perform the following steps to update the FPGA code in the Vantage hardware system:

1. Shut down the host controller. Power off the Vantage hardware system.
2. With both the host controller and the Vantage hardware system powered off, ensure the PCI Express cable between the host controller and the Vantage hardware system is properly connected.
3. Power on the Vantage hardware system. Note that the Vantage hardware system does not turn on at this time; it is powered on after the host controller system powers up.
4. Power on the host controller and observe that both the host controller and the Vantage hardware system are powered up. The fans in the Vantage hardware system are audible when it powers up.
5. Log in to the host controller and launch the MATLAB application.
6. Select the newly installed *Vantage project* folder as the MATLAB working directory.
7. Issue the `activate` command.
8. Issue the `Version` command at the MATLAB command prompt.

- a. Confirm the version number for the Vantage software release.
- b. If the Vantage hardware system reports that updates are needed, follow the command prompts to update the FPGA code in the Vantage hardware system.

As the update proceeds, periodic status reports on the progress are displayed.

Note: It is critically important to wait until this process is completed. Note that some steps can take several minutes. If this process is interrupted for any reason, the Vantage hardware system may be left in an inoperable state.

- c. Once the status reports indicate that the FPGA code updates are complete and a power cycle is required, close MATLAB and shut down your host controller to a completely power-off state.

Note that once the updates are complete, a dialog box is displayed informing the user of the need to perform a complete system power shutdown. During this phase, there may be MATLAB error messages seen while closing the dialog box and shutting down MATLAB. These are normal responses to the FPGA code updates and are not an indication of an actual problem. These responses can be ignored, and a full system shutdown should be completed.

9. Restart the host controller. The Vantage hardware system should power up as well.
10. Start the MATLAB application and run the `Version` command. The command execution should be completed without any error or warning messages.

If there are any issues encountered while implementing the above procedure, contact Verasonics Technical Support (support@verasonics.com) for assistance.

If the `Version` command fails to successfully complete the FPGA updates, perform the following steps for a forced update of all FPGA code:

1. Shut down and restart the system from a full power-off state.
2. Select the desired *Vantage project* folder and issue the `activate` command.
3. Issue the `reprogramHardware` command at the MATLAB command prompt and follow all prompts.

This process can take several minutes.

Once execution of `reprogramHardware` command completes, it will notify that another full shutdown and restart cycle is required.

Note: Once the updates are complete, a dialog box is displayed informing the user of the need to perform a complete system power shutdown. During this phase, there may be MATLAB error messages seen while closing the dialog box and shutting down MATLAB. These are normal responses to the FPGA code updates and not an indication of an actual problem. These responses can be ignored, and a full system shutdown should be completed.

7. System Self-test: VerasonicsVerificationTest Utility

As a last step in preparing the Vantage System for use, run the self-test script (`VerasonicsVerificationTest`) to verify that all digital and analog hardware subsystems are operating within specifications.

Prior to running the test, it is recommended to do a system power shutdown and restart. This ensures that the system is in a known operating state. Use the following steps to perform the self-test:

1. Disconnect all transducers.
2. Open a MATLAB instance.
3. Navigate to the *Vantage project* folder and run the `activate` command.
4. Type `VerasonicsVerificationTest` in the Command Window and press `Enter`.
5. On the GUI window that pops up, click the `System Verification Test` button.
6. Follow and respond to the prompt(s) that appear in the MATLAB Command Window.

The test takes a few minutes, and the final result should indicate "**ALL TESTS PASS**" in the MATLAB Command Window. If any test fails, the "**TEST FAILED**" message appears in the Command Window. The script saves log files in the subfolder *HardwareTest/TestResultLogs*. There are two output files prefixed in the format `VVT_HOSTNAME_YYMMDD_HHMMSS`—one of the files has a suffix `_report.txt` and the other file has the `.log` suffix. The prefix, `YYMMDD_HHMMSS` represents the date and time when the test was started. `HOSTNAME` refers to the hostname of the Host Controller. For example, for a file with the name `VVT_HOST1_2020821_044635.txt`.

- The `HOSTNAME` is `HOST1`
- The test started at 04:46:35am on August 21st, 2020

Send the two output files to support@verasonics.com. In the email, include a description of the problem and the symptoms observed.

8. Software Licensing

The use of Verasonics Software is controlled by the license file—*license.enc* for Vantage System.

The user must update the license file before installing a new software release and after renewing the license.

Starting with release 4.3.1, Verasonics no longer includes a customer license file within the distribution package. In the past, when the user downloaded a distribution from your customer download folder, it included a copy of the organization's license file.

The user can now download the current license file from your *customer download* folder and copy it into the root of your *Vantage project* folder.

All licenses associated with a single organization are contained in a single license file. Consequently, when the user receives a new license file, it must be copied to the *Vantage Project* folders of each of the host controllers whether the host controller is used with the Vantage hardware system or used in simulation-only mode.

Software license file formats

The software license files are distributed in one of two formats—a pre-compiled binary format (*license.enc*) or a plain-text format (*license.txt*). Most of the license files are provided in a pre-compiled binary format; however, some license files are distributed in a plain-text format to satisfy certain regulatory requirements.

Note: Do not edit the pre-compiled license file and the plain-text license, doing so will render them invalid.

8.1.1 Download the license file

The user can find the current version of the license file in their *Customer Download* folder. The *Customer Download* folder is located on the Verasonics download site: <http://downloads.verasonics.com>.

Use your login credentials to access the customer download folder. If you do not have login credentials or if you are unable to access the customer download folder, contact support@verasonics.com for assistance.

8.1.2 Import a binary license file

1. Close any running MATLAB instances.
2. Copy the license file to the root of your *Vantage Project* folder, overwriting any existing license file.
3. Open MATLAB.
4. Set the current directory to the root of the *Vantage Project* folder.
5. To activate the software, open the MATLAB console and type **activate**.

8.1.3 Import a plain-text license file

The following procedure needs to be done once for each *Vantage Project* folder. Once the *license.enc* file is created it does not need to be recreated until a new *license.txt* is downloaded, such as when the license expires, or a new product is added to the license.

1. Close any running MATLAB instances.
2. Copy the plain text license file (*license.txt*) to the root of your *Vantage Project* folder, overwriting any existing license file.
3. Open MATLAB.
4. Set the current directory to the root of the *Vantage Project* folder. Do not execute the `activate` command.
5. Run **importLicenseKey**. The command compiles your plain text license key, creating a *license.enc* file in your *Vantage Project folder*.
6. Run the `activate` command. The Vantage software is now ready to be used.

8.2 Review the Software License

The user can review the contents of their Vantage *software license* by running the `systemInfo` command from the MATLAB command prompt. View the file `systemInfo.txt` in the MATLAB text editor, or any other text editor of your choice. `SystemInfo.txt` describes the licensed features and the date range over which the license is valid.

8.3 Update your Software License

Periodically, the user may need to renew their software license.

Use the following procedure to update the software license.

1. Run the `systemInfo` command from the MATLAB command prompt to generate *systemInfo.txt* in the current directory.
2. Attach the *systemInfo.txt* file to an email message. In the email message, describe the situation briefly and send the email to licensing-request@verasonics.com.

Verasonics notifies the user when a new license file becomes available. Download and install the license by following the instructions described in the previous section.

Note: Please do not request a license without attaching an appropriate *systemInfo.txt* file. This information is required to generate a license.

To renew the current license or obtain a new license for the Vantage System, the above procedure must be executed on the host controller that is connected to the Vantage hardware system.

To renew the current license or obtain a new license for the Vantage Simulator, execute the above steps on the computer used for running the simulator (instead of executing them on the host controller) after completing the software installation.

8.4 Temporary Software License

In case of an emergency, they can contact your local Verasonics representative or Verasonics Technical Support (support@verasonics.com) for a temporary software license.

8.5 Grace Period

All capabilities are licensed for a specific period. In addition, a grace period of thirty calendar days is provided beyond the license expiration date.

Before using the Vantage System for a critical task, it is prudent to check the license expiry date on the capabilities the user intends to use.

8.6 Additional Licenses

The Service Level Agreement with Verasonics entitles the user to additional licenses. Please contact Verasonics for additional information on the number of licenses that the user is entitled to receive.

To update the software license, refer to Section 8.3.

9. Software and Hardware System Compatibility Constraints

The Vantage hardware system is available in several system configurations with optional hardware-level features. The current Vantage software release supports all existing hardware system configurations—this enables the user to run the same Vantage software installation across multiple Vantage hardware systems with different configurations.

On system power-up and initialization, the software checks for the presence of a Vantage hardware system and automatically configures itself for use with the specific hardware configuration associated with the Vantage hardware system.

In general, Verasonics strives to make each new software release backward compatible with all older hardware system configurations and with user acquisition scripts that were developed on older versions of Vantage software and/or Vantage hardware systems. However, there are situations where maintaining backward compatibility can be prohibitively complex or can lead to unacceptable performance compromises. The Verasonics support team provides support to its users in situations where backward compatibility can be restrictive.

9.1 Support for Older Software Releases

Verasonics provides the latest software release to all Vantage System customers in good standing, even if their Vantage Systems are out of warranty and they do not have an active service agreement.

If a user is not using the current software release and requests for support from the Verasonics Support team, the user is mandated to install the current software release before receiving any assistance from the support team.

However, Verasonics recognizes that there are situations where some users may need to continue to work on their ongoing research projects or clinical studies using an older software release and cannot upgrade to the latest software release. Verasonics does provide support and access to older software releases as needed. Contact Verasonics Technical Support (support@verasonics.com) for assistance.

9.2 Support for Obsolete Vantage Hardware Configurations

Systems using P00732-02 Acquisition Module and P00729-03 Backplane hardware module revision levels, which were used only in the earliest Vantage Systems built in 2013 or early 2014, are supported by the 3.0. Vantage software releases but have been declared obsolete and cannot be used with the 3.2.0 or later software release.

If you have a Vantage hardware system containing either of these modules and are interested in updating to the latest software release, a hardware upgrade is required. Contact Verasonics Technical Support (support@verasonics.com) for assistance.

9.3 Support for Older Software Releases on Newer Hardware Configurations

As the Vantage product line continues to evolve, Verasonics frequently releases new hardware-based features and newer versions of hardware modules offering improved performance. In many cases, these newer hardware features cannot be used with older Vantage software releases. In some cases, the older software releases may not function at all.

In situations where it is required to use an older Vantage software release on a newer Vantage Hardware system, contact Verasonics Technical Support (support@verasonics.com) for more information and assistance.

9.4 Support for User Setup Scripts and Verasonics Example Scripts

The acquisition scripts developed for use with an earlier Vantage software release can be used with newer software releases as they become available. Note that this compatibility applies to the "Setupxxx.m" script itself and not to the .mat file it creates.

The Setup script must be re-run on the newer software release to generate a new .mat file that is compatible with this release. The older .mat file is not compatible with the newer software release.

Starting with the 4.3.0 software release, GUI callbacks, and external processing functions can be defined as MATLAB functions in the SetUp script file, eliminating the "text2cell" markup that was used in earlier releases. Releases 4.3.0 and later continue to be backward compatible with the "text2cell" definitions.

Note: With recent MATLAB releases (starting with 2019b), the existence of a persistent variable definition within a `text2cell` function will trigger a MATLAB error. To use current MATLAB releases, any SetUp script using persistent variables must be converted to use MATLAB functions.

For documentation and examples showing how to make this conversion, refer to the folder *Example_Scripts /Vantage_Features /New UI Scheme* in the Vantage software.

10. Best Practices for Managing Software Folders

10.1 Managing Multiple Vantage Software Installations

The user may install any number of copies of the *Vantage Project* folder and locate them wherever they wish on the host controller. To create a duplicate installation, the user can either extract the zip archive again or make a copy of an existing *Vantage Project* folder. The user can choose any name they like for renaming the *Vantage Project* folder directories. The Vantage software does not have any dependencies or references that are linked to the directory name.

Note: It is good practice to include the 3-digit software release ID as part of your project folder name—especially, if the user has multiple Vantage software installations (associated with different software releases) on their Vantage System.

To switch from one Vantage Project folder to another-

1. Close the current MATLAB application.
2. Reopen MATLAB.
3. Select the new *Vantage Project* folder as the MATLAB working directory.
4. Open Command Window, type `activate` command.

It is not required to restart the host controller.

To switch from one version of MATLAB to another-

1. Close the current MATLAB application.
2. Open another version of the MATLAB tool.
3. Select the new *Vantage Project* folder as the MATLAB working directory.
4. Open Command Window, type `activate` command.

To switch from one Vantage software release to another-

When the switches from one Vantage software release to another, and the two software releases require different levels of FPGA code for the Vantage hardware system (when switching between Vantage software release 4.3 and any earlier release), the user will be prompted to allow updates to the Vantage hardware system. After the system updates are completed, a system power cycle is required before using the system.

Note: The procedure is not applicable if Vantage software is running in simulate-only mode and the Vantage hardware system is disconnected or turned off.

10.2 Adding Other Directories to the Vantage Project Folder

The desired *Vantage Project* folder must be selected as the MATLAB working directory. This is because some of the Vantage software functions use relative path definitions that are tied to the *Vantage Project* folder—this constraint is the key element of the design approach to allow multiple software installations on the host controller.

Note: If the user selects a higher-level directory that contains the *Vantage Project* folder, or if the user selects a sub-directory within the *Vantage Project* folder, the Vantage software will not function properly and may cause a MATLAB crash or other problems.

The user can add new subdirectories within the Vantage Project folder. As the `activate` command automatically adds all subfolders within the *Vantage project* folder to the MATLAB path, all folders (subdirectories) are included.

The user can add directories to the MATLAB path that are not included within the selected Vantage project folder. The user can do this by using any of the MATLAB utilities for adding directories to the path. The only constraint is that the user must do this AFTER they have run `activate` command for the current *Vantage project* folder. This is because the `activate` command first removes everything from the MATLAB path (other than MATLAB's own directories), and then adds all directories to the path within the *Vantage project* folder.

10.3 Overview of the Directory Structure within the Vantage Project Folder

Utilities This folder contains many of the MATLAB "helper" functions that can be used within a user's SetUp script directory—such as `computeTXDelays`, `computeTrans`, and `computeTGCWaveform`. These are all ".m" files in the customer release that a user may want to have access to in order to understand system operation or in some cases even modify them. For example, the user may want to edit `computeTrans` to add support for a custom transducer that they have developed.

System This folder contains MATLAB ".p" files and compiled C and mex functions for which source code is not included in the customer release. These files must be present for the system to run, but since most users have no need to ever look at them, we created this sub-folder to get them out of the way in the main folder.

Example Scripts In current Vantage software releases, the Example Scripts have been organized in a multi-level hierarchy of folders to make it easier for you to find the probes and associated scripts that you are interested in.

At the root level of the *Vantage Project* folder, there are two main example script folders, *Examples_NDE_NDT* and *Examples_Biomedical* that divide the scripts into two broad categories of applications supported by the Vantage product line.

At the next level down in *Examples_Biomedical*, there is a separate subfolder for each of the Vantage Hardware system configurations—Vantage 32 LE, 64, 64 LE, and 128-256. Within each of these systems folders, there is a subfolder for each UTA module type supported for that system, and within each UTA folder are folders for each probe supported by that UTA, and within each probe folder, you will find the individual example scripts for that probe.

A fifth subfolder named *Specialty_Applications* can be found under *Examples_Biomedical*. This folder contains folders for a broad range of both specific imaging applications (such as *PhotoAcoustics*) and also examples of how to use specific features of the system (such as *MultiSystemClockSynchronization*). In most of these specialty folders, you will find application notes and other documentation as well as the associated example scripts.

The organization within *Examples_NDE_NDT* is similar to *Examples_Biomedical* but somewhat simpler because of the smaller range of UTA modules and example scripts currently available.

MatFiles The .mat file generated, when you run a SetUp script, is stored in the MatFiles subfolder rather than in the main folder. This helps reduce clutter in the main folder. If you have your own SetUp scripts that you have developed or created by modifying example scripts from earlier releases, you should modify them to store the results in the MatFiles subfolder. All that is required is to update the line with the `save` command to include the MatFiles path.

For example, in a SetUp script that creates a file named "*myScript.mat*", you should make the following change to the `save` command in the script to use the MatFiles subfolder.

Previous `save` command: `save('myScript')`

New `save` command: `save('MatFiles/myScript')`

Note that no change is required to any function that reads or loads a .mat file, since the *MatFiles* folder will always be added to the MATLAB path by `activate` command.

Note: When using the "*MatFiles*" directory, ensure that duplicate .mat files are not created in multiple subdirectories or in the main *Vantage project* folder. If this happens, MATLAB will use the *first* .mat file it finds in its search path, which may not be the one intended. For example, if there is a copy of "*myScript.mat*" in the main *Vantage project* folder, and the script is modified such that the new .mat file output is stored in the

MatFiles folder, VSX would use the old version in the *Vantage project* folder. This occurs because the *Vantage project* folder appears earlier in the MATLAB search path compared to the *MatFiles* folder. The best way to avoid such an issue is to delete any .mat files that are inadvertently created in the root level of the *Vantage project* folder.

11. System Features

11.1 TPC Profile 5 High Voltage Monitor

All Vantage System configurations and software releases have provided a built-in fault detection monitoring feature for the transmit HV supply when using imaging TPC provides 1 - 4. If the measured actual HV supply voltage (measured by a monitoring circuit that is independent of the control of the HV) differs from the programmed voltage by more than 5 Volts, a fault condition is reported by the hardware system, and execution of the event sequence is terminated. The monitor comparison is done once per acquisition event, using the measured voltage immediately before the start of the event so the measured value will not be affected by droop or transients that occur during and immediately after the active transmit interval. Starting with the 3.1 release, a similar monitoring function is now available for use with TPC Profile 5 (in all prior releases the monitoring feature was disabled for transmit events using Profile 5). The profile 5 monitor functionality is exactly the same as described above for profiles 1 - 4, with the exception that the fault detection thresholds are user-programmable. The function "setProfile5VoltageLimits" provides control of this feature through the following syntax:

```
rc = setProfile5VoltageLimits(P5HVmin, P5HVmax);
```

The arguments P5HVmin and P5HVmax are MATLAB doubles that specify the lower and upper thresholds at which an error will be triggered, in units of Volts over the range from 0 to 100 Volts. The returned value "rc" will contain a MATLAB string, set to the value 'Success' if the command completed successfully, or some other string identifying the problem if there was an error. The profile 5 monitoring feature can be disabled by setting any pair of values such that P5HVmin is equal to or greater than P5HVmax. The profile 5 monitor function is also disabled by default when starting to run a script with VSX. You must invoke the setProfile5VoltageLimits command after starting the script, to enable monitoring.

Note that you must manually update the monitor thresholds whenever you change the TPC Profile 5 HV setting (one approach would be to disable the monitor prior to changing HV, and then re-enable it with new limits after the voltage has actually settled at the new setting. An alternative would be to change the limits, so they include both the old and new settings while the voltage is in transition, and then change again to new limits for the new setting after the transition is complete.)

Note also that due to component tolerances, there will always be some difference between the programmed values and the actual voltages in the hardware system, both in the HV control setting and in the profile 5 monitoring thresholds. As a result, you should always allow at least several tenths of a Volt margin between the monitor limits and the actual HV setting. Using the example script described below, you can experiment with moving the limits in 0.1 Volt steps closer to the actual voltage setting, to determine at what level the limit threshold will actually be reached on your system.

The example script "SetUpL11_5vFlashPushP5Mon" illustrates the use of this feature, with GUI controls on the min and max limits. Comments in the script provide further details on the use of this feature. This script is located in the folder *ExampleScripts / Vantage_Features / Basic_Features*.

11.2 RF Data Acquisition Time Tag

This is a user-programmable feature. When enabled, the time tag feature will overwrite the first two samples of each receive data acquisition event with a 32 bit "time tag" value taken from a free-running counter in the hardware system. As a result, the RF data stored in memory from every acquisition event will include a time tag identifying the precise time at which it was acquired relative to other acquisition events with a resolution of 25 μ sec and a range greater than 24 hours. If desired, you can also associate the relative time tag values with an absolute time and date taken from the host controller time/date functions.

When a script starts running the time tag feature will be disabled by default. It can be enabled or disabled whenever desired by using the command "Hardware.enableAcquisitionTimeTagging()" with the following syntax:

```
boolean result = Hardware.enableAcquisitionTimeTagging(enable);
```

The argument "enable" is a boolean value set to 'false' to disable time tags or 'true' to enable them. The returned value "result" will be a boolean variable, set to 'true' if the command was completed successfully, or 'false' if there was an error.

The enabled status is "global", in the sense that while time tags are enabled, they will be added to every RF data acquisition event executed by the HW system, and they will be applied to every receive channel included in the DMA transfer to the Receive Buffer in the host computer. There is no mechanism for applying the time tag selectively to individual channels. The Recon function is not aware of the presence of time tags, so if the Recon processing makes access to the first two samples from a receive event, the time tag values will be used as if they were legitimate RF data. In most acquisition scripts this is not a problem since the first two samples will come from before or during the transmit activity, and thus the actual sample values are typically meaningless anyway. If it was necessary to prevent the time tags from corrupting the receive data samples you could either arrange the acquisition timing so the first two samples would not be used for RF processing, or you could use an external processing function to force them to zero after the time tag value had been read.

Enabling and disabling the time tag feature will not affect the state of the 32-bit time tag counter. It is reset to the zero state at system power-up and then free runs continuously as long as the system is powered up. Since it increments every 25 usec, the 32-bit value will wrap around back to zero after approximately 107,000 seconds or 29.8 hours. If desired, you can also program the counter to wrap back to zero after exactly 24 hours or 86,400 seconds. You can also reset the counter back to zero whenever desired. Both of these options are controlled through a separate mex function, "setTimeTaggingAttributes", with the following syntax:

```
result = setTimeTaggingAttributes(wrap24, reset);
```

The argument "wrap24" must be a MATLAB string, set to either '1' or 'true' to enable the 24-hour wraparound, or '0' or 'false' to allow the full 32-bit count wraparound. The "reset" argument will force the counter value back to zero if it is set to '1' or 'true', at the point in time when the command is executed. The "reset" argument will do nothing if it is set to '0' or 'false'.

To map the time tag counter value to an absolute time and date, simply record the time and date from the host computer using an appropriate MATLAB function, immediately after sending the "reset" command to the time tag counter. Then subsequent time tag values can be interpreted as an offset from that recorded time.

The example script "SetupL11_4vFlashTimeTag" provides an example of enabling and disabling the time tag feature as well as extracting the time tag value in an external processing function and displaying it. The utility function "ShowTimeTag" (in the Tools folder) provides an example of extracting time tag values from the receive data buffer in the MATLAB workspace.

12. Maintenance

The Vantage hardware system does not have any user-serviceable parts inside the chassis. DO NOT open the side covers of the Vantage hardware system. The anti-tamper stickers placed on the Vantage hardware system provide visible evidence of unauthorized removal of the side covers—this can result in the termination of the system warranty.

The only parts that can be replaced are the fuses located in the AC Inlet on the back side of the Vantage hardware system.

12.1 Fuse Replacement

The two fuses in the AC Inlet on the back of the Vantage hardware system can be replaced with 10A 250VA, Slo Blo, Bussman MDA Series Fuses.

12.2 Fuse Replacement Procedure

Use the following procedure to replace a fuse:

1. Power off the Vantage hardware system chassis and unplug the power cord.
2. Locate the fuse door at the rear side of the Vantage chassis.
3. Open the fuse door using the tool provided.



Figure 9: Fuse door at the rear side of the Vantage chassis

4. Gently remove the fuse housing from the receptacle using the tool.

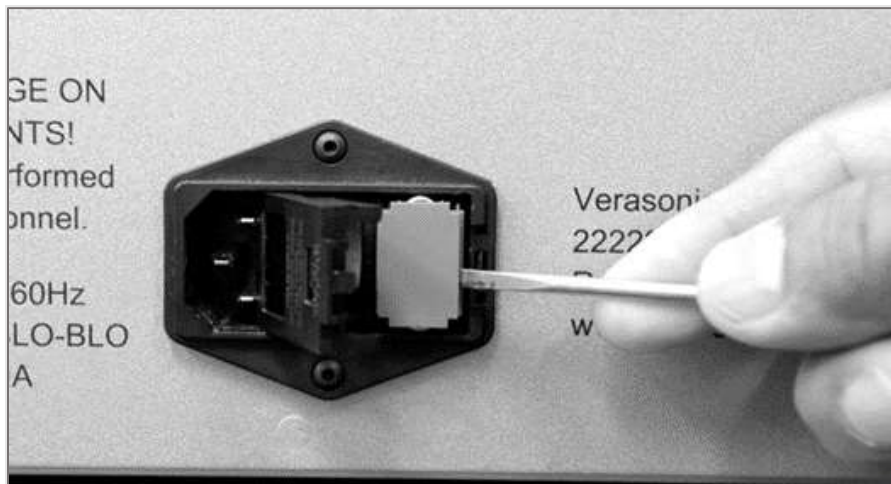


Figure 10: Fuse housing removed from the receptacle

5. Remove both fuses from the fuse holder and replace them with the replacement fuses provided.

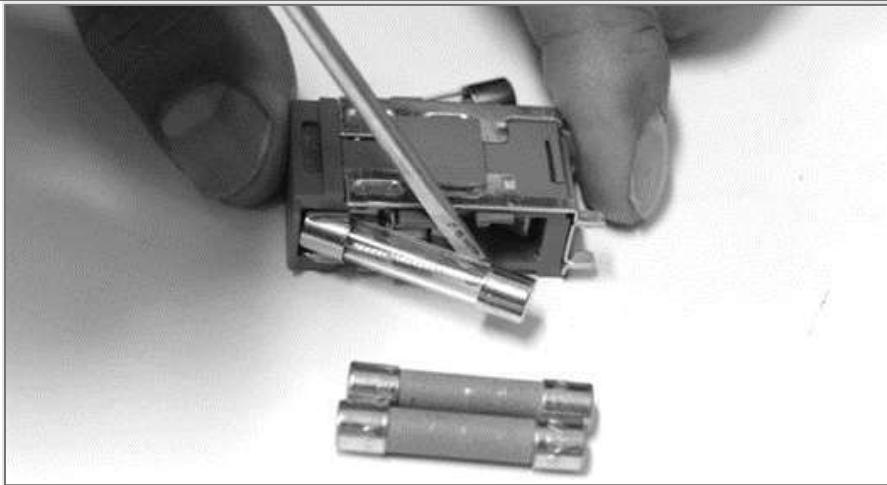


Figure 11: Replacement fuses placed in the fuse holder

6. After replacing both the fuses, gently install the fuse holder in the Vantage chassis receptacle.



Figure 12: Fuse holder installed in the receptacle

7. Close the fuse door and discard old fuses.

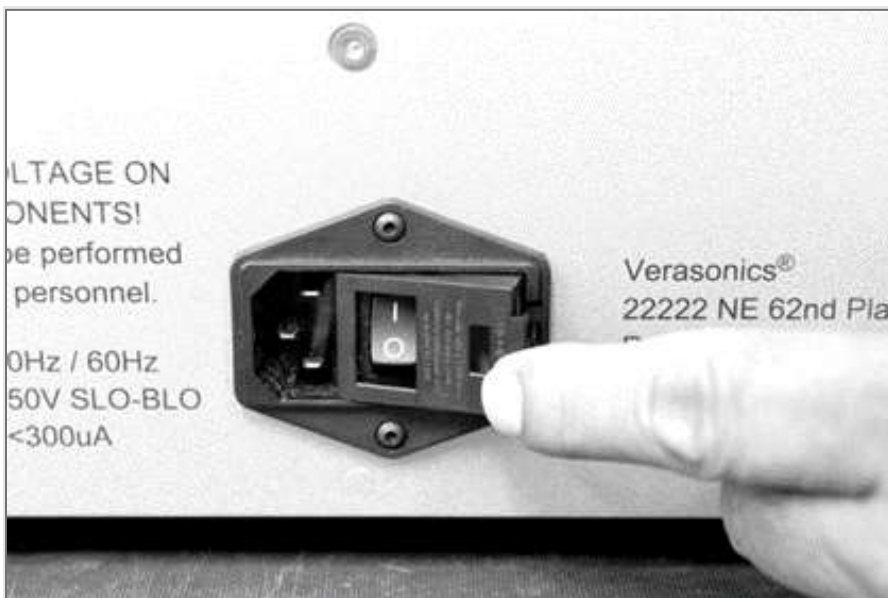


Figure 13: Fuse door closed after replacing the fuse

13. Licensable Features

The Verasonics Software license provided with your Vantage System enables all the capabilities for which a license has been acquired. The complete list of capabilities that can be licensed in the current Vantage software release is shown below:

Hardware License

A Hardware license enables the use of the Vantage software with one or more Vantage hardware systems.

Simulation License

A Simulation license enables the use of the Vantage software when it is disconnected from a Vantage hardware system.

ArbwaveTk License

An ArbwaveTk license enables the use of the Arbitrary Waveform Toolkit, a subsystem of the Vantage software. With an ArbwaveTk license, the user can run the "*Arb_Main*" utility within the ArbwaveToolbox software package provided with Vantage software releases. This is a standalone, self-contained software utility used for the generation and analysis of arbitrary waveforms.

ArbwaveTx License

An *ArbwaveTx* license enables the generation of custom-designed waveforms when the Vantage software is used in conjunction with a Vantage hardware system. When this capability is enabled, the Vantage hardware system can transmit any waveform it can generate. When this capability is disabled, the Vantage hardware system can only transmit waveforms that can be produced with the parametric TW.type.

ExtendedTx License

An ExtendedTx license enables the generation of extended transmit waveforms when the Vantage software is used in conjunction with a Vantage hardware system.

This license permits generating "push" transmit waveforms using TPC Profile 5 with the internal auxiliary TPC power supply on the Vantage hardware system. This capability has no counterpart in simulation mode, so the presence/absence of this capability has no effect on the Vantage software simulation functionality.

Trigger License

A Trigger license enables the use of the Trigger In and Trigger Out capabilities of the Vantage hardware system when it is connected to the Verasonics Vantage software. Trigger capability is an optional feature on Vantage 64 and Vantage 64 LE models but is automatically licensed as a standard feature on Vantage 128 and Vantage 256 models.

Recon License

The Recon license enables the use of the "*Reconstruction processing*" function within the Vantage software. Recon capability is an optional feature on the Vantage 64 System but is automatically licensed as a standard feature on all other Vantage System models.

The following section is a French translation of the Safety and Regulatory Guidelines section.

Appendix A. Sécurité et Conseils Réglementaires

A.1 Explication des symboles

Safety Symbol	Meaning	Description
	"Voir le Manuel de l'Utilisateur"	(Ce Document)
	"Attention"	Ce symbole indique une situation potentiellement dangereuse qui peut entraîner des blessures mineures ou modérées. Le Manuel de l'Utilisateur (ce document), et en particulier les consignes de sécurité suivants, doit être consulté quand ce symbole est en évidence.
	"Tensions Dangereuses"	Ce symbole indique la présence de tensions dangereuses à l'intérieur du couvercle, et qu'il n'y a aucun composant réparable par l'utilisateur. Ne pas enlever les panneaux latéraux!

A.2 Mesures de Sécurité



+**Opérateur: Il est important que vous suiviez ces exigences de sécurité +pour vous protéger et pour la protection du système

+Verasonics. Veuillez exiger a tout nouvel utilisateur du système de se +familiariser avec les consignes de sécurité dans ce manuel.**

	NOTE: Le système Verasonics n'est pas certifié pour l'utilisation avec des sujets humains. Pour plus de détails sur les risques et sur les prévisions de sécurité du système, et pour autres informations sur les précautions et préparations à prendre afin d'utiliser le système dans le cadre de la recherche avec des sujets humains, veuillez contacter Verasonics: Support@Verasonics.com .
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Après l'installation initiale du système (et après toute installation de logiciel), lancez la routine d'auto-test (à la commande Matlab, tapez "VerasonicsVerificationTest"). Assurez-vous d'abord qu'aucune sonde ne soit branchée. Veuillez prendre contact avec "Verasonics Support" si le test ne fonctionne pas, ou si le test indique un problème. Veuillez répéter le test après transport du système, après avoir débranché / rebranché le câblage du système ou autre démontage partiel, et après installation de logiciel etc.

Ne jamais faire fonctionner le système sans panneaux en place. Les panneaux sont nécessaires pour le refroidissement du système, le blindage CEM, et la protection de l'utilisateur et du système. Des tensions potentiellement dangereuses existent à l'intérieur du boîtier. Aucune pièce intérieure n'est réparable par l'utilisateur. En cas de panne, indiquée par l'auto-test, ou par analyse de résultats, le système doit être livré à Verasonics pour mise à niveau ou tout autre entretien. Preuve de l'enlèvement non autorisé des couvercles latéraux peut entraîner l'annulement de la garantie du système.

Ne jamais stocker les liquides sur le système ou à proximité, et surtout ne jamais renverser de liquide sur ou dans l'enceinte pour protéger l'utilisateur et le système d'une défaillance électrique. Si un déversement se

produit, éteindre immédiatement le système (ordinateur et Vantage hardware system inclus) et contacter Support@Verasonics.com. Assurez-vous que des mesures prises pour protéger le système contre les déversements ne restreignent pas aussi le flux d'air pour le refroidissement. Certaines sondes prototypes peuvent être mal blindées et peuvent interférer avec d'autres instruments, et en présence de dispositifs médicaux sensibles aux rayonnements RF, des soins supplémentaires doivent être pris. Si le système sera utilisé dans un environnement où cela pourrait gêner, le client doit effectuer des tests CEM de la configuration complète du système (sonde, câblage, Vantage hardware system, ordinateur, et tous les autres appareils connectés). Les essais CEM des composantes individuelles ne sont pas suffisants pour estimer la performance EMC du système global.

Sécurité et alimentation du système complet: En usage normal, l'appareil Vantage hardware system doit être interconnecté avec un ordinateur, un écran, et potentiellement d'autres dispositifs qui peuvent chacun avoir leur propre connexion d'alimentation. L'utilisateur doit être conscient du fait que le courant de fuite du à l'ensemble de ces dispositifs interconnectés peut dépasser les limites de sécurité. Autres problèmes avec les connexions à l'alimentation, comme les connexions à terre inadéquates, peuvent entraîner des risques de sécurité potentiellement graves à l'opérateur ou aux sujets de l'essai. Il est recommandé que l'utilisateur effectu des mesures de sécurité de ligne, par exemple pour l'isolation, courant de fuite, et impédance de la connexion à terre. Ces vérifications de sécurité de l'alimentation doivent être répétées chaque fois que le système est démonté et remonté, ou soumis à des transports rugueux.

Rappel: Verasonics recommande à l'utilisateur de lancer l'auto-test (VerasonicsVerificationTest) après toutes modifications, comme par exemple, mis à jour (logiciel ou matériel), transporté, ou démonté et remonté. Pour exécuter le test, démarrez le système, lancez Matlab, tapez "VerasonicsVerificationTest" et répondez aux questions éventuelles. Le test devrait terminer avec un message de success (Pass) ou erreur (Fail). En cas d'erreur, veuillez transmettre le fichier créé par le test à support@verasonics.com.

A.3 Dégâts de transports et livraison

Après toutes livraisons veuillez examiner soigneusement le carton d'expédition pour tout signe d'endommagement. Examinez également les capteurs de choc attachés à l'extérieur du carton. Veuillez nous communiquer au plus tôt si l'un d'entre eux soit déclenché, ou s'il y a d'autres indications de dommages ou de mauvais traitement en cours d'envoi. Après avoir déballé le système, inspecter soigneusement le boîtier du Vantage hardware system et ses connecteurs afin de vérifier que la livraison n'aurait pas abîmée le système.

A.4 Immunité CEM (Sensibilité au bruit électromagnétique)

Si l'image ultrasonore apparaît contaminé par du bruit, il est possible que le système ne soit pas suffisamment blindé contre le rayonnement ou la conduction d'énergie provenant de l'environnement électromagnétique. Dans ce cas, veuillez contacter Verasonics pour des conseils supplémentaires:

support@verasonics.com.

A.5 Mise à la terre électrique

Lors de l'installation, assurez-vous que le circuit d'alimentation soit fourni d'une connexion à la terre fiable. Assurez-vous aussi d'utiliser le câble d'alimentation comprenant un connecteur à trois broches (avec terre) fourni par Verasonics

A.6 Transport du Système

Prenez soin en soulevant le système Vantage, qui peut peser jusqu'à 20 kg (44 lb). Si le modèle est capable d'émission HIFU, le boîtier peut peser jusqu'à 26 kg (57 lb). Lors de déplacement, veuillez toujours utiliser le carton d'emballage fourni par Verasonics, conçu pour la protection du système contre les chocs et abus de transport. Le carton est muni de poignées pour faciliter le soulèvement. Rappelez-vous de plier les genoux!

A.7 Conseils à l'utilisateur

- Si l'installation du logiciel est nécessaire, suivez les instructions d'installation soigneusement et précisément, en faisant particulièrement attention au redémarrage automatique pour assurer la configuration correcte du logiciel. L'installation du logiciel sera corrompue si vous relancez manuellement avant que les redémarrages automatiques puissent se faire. Utilisez le système uniquement à des températures ambiantes comprises entre 50 à 95°F (10 à 35 °C)
- Assurez-vous que la performance de la ligne d'alimentation soit entre les limites spécifiées pour: la tension, la fréquence, le niveau de puissance disponible, ainsi que les niveaux de surtensions, les baisses de tension, et les décrochements.
- Ne bouchez pas les grilles de ventilation qui protègent le système contre la surchauffe. Si les ventilateurs ne fonctionnent pas lorsque le système et l'ordinateur sont en marche, veuillez éteindre le système et prendre contact avec Verasonics au plus tôt.
- Auto-test du système: S'il y a jamais question de savoir si le système fonctionne toujours correctement, éteignez l'ordinateur et aussi l'interrupteur de l'alimentation du VDAS. Après quelques secondes, d'abord allumez l'interrupteur du Vantage hardware system, et ensuite redémarrez l'ordinateur. Lors du redémarrage et le relancement de MATLAB, débranchez tous les transducteurs et lancez le programme d'auto-test Verasonics "VerasonicsVerificationTest", et vérifiez que le système passe sans aucune erreur. En cas d'erreur, ou d'avis d'échec apparaissant dans la fenêtre de commande, contactez support@verasonics.com, et prenez soin d'envoyer les résultats du test.
- Si l'utilisateur a l'intention de contrôler la sortie acoustique du système, le programme VerasonicsVerificationTest doit être exécuté à chaque démarrage afin d'assurer que le système fonctionne selon ses spécifications. Les mesures de sortie acoustiques, ou d'autres tests de vérification effectuées sur le système, doivent être vérifiées après chaque modification du logiciel (software) ou du matériel (hardware), de mise à niveau, ou de réparation ou maintenance.
- Veuillez conserver le carton d'expédition et les matériaux d'emballage; ils ont été spécialement conçus pour protéger le système en cas de transport. Si jamais vous avez besoin d'expédier le système de nouveau à Verasonics ou à un autre endroit, l'emballage d'origine doit être utilisé.

Fin de la Version Française — End of French Document

Appendix B. Guide for Use of the UTA System Configuration

The UTA configuration for the Vantage hardware system provides user-replaceable modules on the front of the system and allows use with a variety of transducer connector types and pinouts. This makes the Vantage System compatible with different families of commercially available ultrasound probes as well as Verasonics-provided probes or custom probes developed by a Vantage customer. By purchasing multiple UTA adapter modules, a single Vantage System can easily be reconfigured for use with more than one probe/connector family.

The UTA configuration was introduced in September 2015; Vantage Systems built prior to this date do not have the UTA feature and instead use a "captive SHI" on the front of the system that only provides the HDI-format 260-pin connector (equivalent to UTA 260-S or UTA 260-D).

All Vantage software releases provide full support for the new features of the Vantage UTA hardware configuration, while also maintaining full compatibility with the earlier pre-UTA "captive SHI" systems.

B.1 Adapter Modules Supported by the System

Refer to the [Vantage Product Specification](#) document (located in the *Vantage project/Documentation* folder) for a listing of all UTA modules that are currently available for use with the system and a brief description of specific features and compatibility restrictions associated with each of the UTA modules.

B.2 UTA-related Terminology and Definitions

To make sense of Verasonics documentation and user instructions, we have defined new terminology and hardware module names for the components of a UTA system, while remaining consistent with the terminology that was used in pre-UTA systems. In some contexts, these names can be ambiguous and confusing; the definitions given below aim to resolve that confusion.

UTA: This is our name for the overall system hardware configuration with the UTA feature installed, such as "Vantage 256 UTA", while the earlier or pre-UTA HW configuration would simply be identified as "Vantage 256". The physical configuration of these two systems is as follows: A Vantage 256 system has the SHI module installed at the front of the system chassis, while a Vantage 256 UTA system has the UTA baseboard module installed in place of the SHI and then a UTA adapter module is connected to the front of the UTA baseboard.

SHI module: This is the physical hardware module installed at the front of a pre-UTA system. SHI is an acronym for "ScanHead Interface". The SHI module includes the HDI-format probe connector accessible at the front of the system (or two HDI-format connectors on a Vantage 256 system). A Vantage 128 pre-UTA system is functionally identical to a Vantage 128 UTA system with the UTA 260-S adapter installed. Similarly, a Vantage 256 pre-UTA system is functionally identical to a Vantage 256 UTA system with the UTA 260-D adapter installed.

UTA Baseboard: This is the functional equivalent of the SHI module for UTA systems. Instead of providing probe connectors, the UTA Baseboard provides the mechanical and electrical interface to support a UTA Module.

UTA Module: This is the user-replaceable module that attaches to the front of a UTA system and provides the physical probe connector(s).

B.3 Removing and Replacing UTA Modules

The UTA modules are intended to be easily and quickly removed and replaced without the use of tools, but the following guidelines must be followed to ensure reliable performance and prevent damage to the system:

The Vantage hardware system should be powered down before removing or replacing a UTA module. If the system is powered on while a UTA module is only partially connected, it could damage the hardware system. In addition, full system reinitialization is required for the system software to interact correctly with the newly installed UTA module. To help enforce this requirement, the Vantage hardware system goes into a fault condition if it senses a UTA module being removed or replaced while the power is on, or if the system is powered up with no UTA module installed. This fault state is clearly identified the 4 LEDs on the front of the

system flashing continuously. To clear this fault condition, the user must shut the system down to a full power-off state, then finish installing the desired UTA module, and then restart the system.

To remove a UTA module, use the following procedure:

1. First shut down the system to a power-off state, and then loosen all four thumbscrews on the front of the module.
2. Gently pull on the module handles simultaneously from both sides, while striving to keep the face of the module parallel to the front of the system. To prevent damage to the small and fragile connectors on the rear of the module, place them in a suitable container or flat on a table so that nothing comes in contact with the connectors.

To install a UTA module, use the following procedure:

1. While installing a UTA module ensure that the Vantage System powered off.
2. Pick up the module from both side handles, and make sure you do not have it upside down—the Verasonics label printed on the module should be at the top (there is also mechanical keying in the corners, that prevents a UTA module from being fully installed if it is upside down).
3. Align the module as closely as possible with the opening on the front of the system and parallel with the face of the system and gently push it into place until you can engage all four thumbscrews.
4. Tighten all four screws until you feel some resistance, and then continue to tighten a bit at a time on each screw to preserve the parallel orientation. Continue to tighten the screws until the module is fully seated onto the system—the gap between the module and the face of the system should be very small.

Do not tighten the thumbscrews any more than you can easily do with your fingers.

Do not use the UTA module handles to lift or move the system. This could damage the module. The handles are only to be used for removing and replacing the UTA modules. To lift or move the system always hold it from the bottom.

B.4 Writing User Setup Scripts for Use with the UTA System

All requirements and guidelines for writing SetUp scripts for use with a UTA system configuration are identical to the pre-UTA configuration, with one exception: the script must identify the connector type that is required, by specifying a new field in the Trans structure- *Trans.connType*. In all earlier Vantage SW releases for use on pre-UTA hardware systems there was no need to identify the connector type since only one connector type existed that was identical on all system configurations.

When a SetUp script is run with the Vantage software and a hardware system is present, VSX will determine which connector type is present on the hardware system based on the type of UTA adapter module that is installed. If the system connector type does not match the connector type required by the script, the software will exit with an error condition (or can be directed to revert to simulation-only operation).

Trans.connType Definition

This is a new field that has been added to the Trans structure to support the UTA feature. *Trans.connType* is a required field for the Vantage System SW; it is a MATLAB double set to an integer value used as an index to identify one of the connector types supported by the UTA system. The following connector type index values are recognized by the Vantage software:

Trans.connType = 1 identifies the HDI-format connector, identical to the probe interface used on pre-UTA Vantage Systems. For backward compatibility with the scripts developed with earlier SW releases, VSX creates *Trans.connType* with a default value of 1 if it was not defined in the user SetUp script (but only if the value of *Resource.Parameters.numTransmit* matches the requirements of the HDI connector, i.e. either 128, or 256 with *Resource.Parameters.Connector* = [1 2]). This connector type is used on the UTA 260-S, UTA 260-D, and UTA 260-MUX adapter modules.

Trans.connType = 2 identifies the interface from either a “break-out board” UTA adapter (providing direct access to the element signals, with no specific probe connector at all), or a custom UTA adapter built by a user.

`Trans.connType = 3` identifies the MS family connector interface and pinout (Cannon DL-360 connector with 256 element signals). This connector type is used on the UTA-360 adapter module.

`Trans.connType = 4` identifies a connector interface developed by Verasonics (256 element signals using an ITT Cannon DLP408R 408 pin ZIF connector). This connector type is used on the UTA 408 adapter module.

`Trans.connType = 6` identifies the Hypertac 160 pin, 128 channel connector used by commercially available probes for NDT applications. This connector type is used on early versions of the UTA 160-DH/32 LEMO adapter module where disconnect sensing is provided for the connector.

`Trans.connType = 7` identifies the GE "D" series probe family connector interface. This connector type is used on the UTA 408-GE adapter module.

`Trans.connType = 8` identifies the "direct connect" interface to the probe, designed to mate directly with cable termination "paddle boards" at the end of the probe cable assembly. This connector type is used on the UTA 256-Direct and UTA 1024-MUX adapter modules.

`Trans.connType = 9` identifies the Ultrasonix format 156 pin, 128 channel connector. This connector type is used on the UTA 156-U adapter module.

`Trans.connType = 10` identifies the Ipx 160 pin, 128 channel connector. This connector type is used on the UTA 160-SI/8 LEMO adapter module.

`Trans.connType = 11` identifies the single-element Lemo coax connector. This connector type is used on the UTA 64-LEMO and UTA 128-LEMO adapter modules, and also on the other NDE/NDT adapter modules using the Hypertac or Ipx connectors.

`Trans.connType = 12` identifies the Hypertac 160 pin, 128 channel connector used by commercially available probes for NDT applications. This connector type is used on current production versions of the UTA 160-DH/32 LEMO and UTA 160-SH/8 LEMO adapter modules, with no disconnect sensing for the Hypertac connector.

There are two other special `Trans.connType` values that can be defined in a SetUp script:

`Trans.connType = 0` identifies a simulation-only script that is not intended to be used on a Vantage hardware system at all, and thus there is no need to identify a probe connector interface. When this value is present in the script, the system will operate in simulation mode only (and will ignore the configuration of the hardware system if one is present). None of the system HW compatibility constraints will be applied (for example, the number of transmit and receive channels can be set to any desired value in the range from 1 to 256). Setting `Trans.connType` to 0 has the same effect as setting `Resource.Parameters.simulateMode` to 1 in earlier software releases (if a user script has set `Trans.connType` to 0 and `Resource.Parameters.simulateMode` is undefined, VSX will set a default value of 1 for `simulateMode`). VSX will assign a default `Trans.connType` value of zero for any script that has not defined it but has set `Resource.Parameters.numTransmit` to a value incompatible with the HDI connector; the result will be that the script will only run in simulation mode.

`Trans.connType = -1` is for use in test scripts that are intended to be compatible with all UTA adapters. When VSX initializes a script with this value for use on a HW system, the -1 will be replaced with the index value representing the connector type that is actually present on the UTA adapter attached to the system. This allows test scripts or other utilities to be written in a 'generic' format that will automatically adapt to the actual system hardware configuration when they are run.

B.5 Element to Channel Mapping

The `Trans.Connector` array (or the selected column of `Trans.HVMux.Aperture` for HV Mux probes) defines the mapping from individual transducer elements to the associated channel in the Vantage hardware system for the selected connector or connectors. These channels are identified in the [Sequence Programming Manual](#) as "I/O channels", to distinguish them from physical hardware channels (in some cases, such as when using a single HDI connector on a Vantage 256 system, the number of I/O channels may be less than the number of physical hardware channels). There is an exact 1-to-1 mapping between I/O channels and

columns of the Receive Data buffer in the MATLAB workspace; the number of columns in the Receive buffer definition (`Resource.RcvBuffer.colsPerFrame`) will always be equal to the number of I/O channels for the selected connector(s).

In summary, the length of the `Trans.Connector` array will always be equal to the number of transducer elements identified in the `Trans` structure. The `N`th entry in that array will define the connection for element `N` of the transducer definition, and the value of that entry will identify the I/O channel number the element is connected to. A value of zero means that element is not connected at all and cannot be used by the system. All of this remains exactly the same for the UTA system configuration, for all UTA adapter modules.

Defining the `Trans.Connector` array for a user-supplied transducer

If you are defining a script for a probe recognized by the Vantage software and supported through the `computeTrans` function, all of the element-to-channel mapping is done for you and most of the time you don't even have to think about it. But if you need to connect test equipment to a particular individual element signal at the probe connector, you will need to find the associated connector pin number (the `Trans.Connector` array defines the element-to-channel mapping but does not specify which physical connector pin is associated with each element). Even worse, if you are connecting a transducer that is not supported by `computeTrans` to the system, you will have to figure out how to define the `Trans.Connector` array. These issues were complicated enough on the original system; in the UTA system they get even more confusing since multiple UTAdapters are supporting multiple connector types to choose from, and each UTA adapter and connector has a unique pin assignment at the connector and a unique mapping of those pins to system channels.

The Vantage software includes some utilities to make the element-to-channel-to-connector pin mapping easier to understand:

showEL After running any script with VSX, and then exiting VSX, the MATLAB workspace will still contain all of the structures that were used while the script was running. Type the command `showEL` at the command prompt, and a listing of all transducer elements is displayed in the MATLAB Command Window, showing the system I/O channel number, physical connector pin number, and Apod array entry number (applies to both `Receive.Apod` and `TX.Apod`) associated with that element. Enter the command with an argument, i.e. `showEL(7)`, and the mapping only for the identified element will be displayed. While working with a script, this utility makes it quick and easy to find the connector pin number and I/O channel number (which is also the column number in the Receive Buffer for the associated receive data) for that element.

showCH and showAPOD These utilities are identical to `showEL` as described above and will display the same information, except that the listing will be sorted by I/O channel numbers or Apod array entries instead of by elements.

To define the `Trans.Connector` array for a probe that is not recognized by `computeTrans`, you must first determine from the probe design documentation which physical connector pin each element connects to. Then you can use one of the utilities described above to find the mapping from each connector pin to system I/O channel, so you can put the correct channel number in each entry of the `Trans.Connector` array. Refer to the [Sequence Programming Manual](#) for additional information on integrating custom transducers into the system.

Appendix C. Verasonics Scripts

Vantage software includes many example programs ("m-scripts") written in the Mathworks MATLAB programming language that specify, through a set of MATLAB structures, all ultrasound parameters controlling ultrasound acquisition as well as data processing and image display.

By convention, Verasonics names the ultrasound parameter scripts "Setup_ProbeName_ScanType.m" (e.g. "SetupL11_4vFlash"); such parameter initialization scripts are referred to as "Setup" scripts.

When executed, these Setup scripts call several Verasonics MATLAB helper functions and compiled C-language files, which MATLAB calls ".mex" files. Verasonics mex files use the Verasonics HAL to communicate with the Vantage hardware system. The HAL is a low-level software library that permits controlling the Vantage hardware system with high-level programming.

The Vantage project folder is the top level of the Vantage software hierarchy. Within this folder are m-scripts and mex files that the user sees in their MATLAB file structure. Programs and functions use the MATLAB environment to specify and run ultrasound imaging sequences on the Vantage hardware system. The master Verasonics script is *VSX*, the "Verasonics Script eXecutor", which loads the user's specifications, sets up the graphics controls and display in MATLAB, and calls the runtime program "runAcq" (a compiled mex file) that interacts with the HAL and the hardware. *VSX* usually runs in a continuous loop, until the user quits by closing the graphical user control panel.

To execute a Setup script-

1. Launch MATLAB and navigate to your Vantage project folder.
2. Activate the Vantage project folder by issuing the `activate` command.
3. Execute one of the sample *Setup* scripts by navigating to the folder containing the desired script. Right-click on the script of interest and select the *Run* option. This executes the script and creates a *.mat* output file; these *.mat* files store the ultrasound parameters that were created by the *Setup* script.
4. Note that *VSX* asks the user for the name of the *.mat* file that contains their desired settings.
The example *Setup* scripts are provided in the *Examples_Biomedical* and *Examples_NDE_NDT* folders. These folders should not be modified. Instead, we recommend that the user begins the development of a custom script by copying an example script into a new subfolder and editing that copied file. This way, the user can have access to the unmodified original folder in its original location.
5. Run the Vantage software from within an activated Vantage project folder. When a Vantage project folder has been activated, it becomes the current directory in MATLAB. This is required for the Vantage software to operate properly.

The user can add other directories to their MATLAB path. These do not have to be located within the Vantage project folder. (Note, however, that directories outside the Vantage project folder are removed from the path by the `activate` command. Any such directories must be added to the path after running `activate` command.)

C.1 Starting an Imaging Script

At the MATLAB prompt, enter *vsx*, which is case-sensitive. *vsx* then prompts the user for the name of the ".mat" file that they want to use. For example, enter "L11-4vFlash" (do not include the ".mat" extension) to use the *.mat* file created by the *SetupL11_4vFlash.m* example script.

After a brief pause, a GUI control window and an image display window appear. If the Vantage hardware system and host controller are correctly connected and powered, Vantage software is installed, and with an L11-4v probe connected, the image display window displays live imaging data.

Note that the user can still run the script without connecting it to a transducer and verify that the system is functioning. When a transducer is not connected to the system, the image display window displays low-level thermal noise instead of displaying live imaging data.

To build a script-

1. Use the MATLAB "Current Folder" hierarchical viewer to find the desired script, such as "ExampleScripts/L11- 4v/SetupL11_4vFlash.m".

2. Right-click the file and choose "Run" from the contextual menu. This creates "L11-4vFlash.mat" and saves it in the MatFiles sub-folder. The user can then run VSX (case-sensitive) and type "L11-4vFlash" to load and run that script.
3. If the Vantage hardware system is not detected by the HAL, the system drops into a simulation mode in which simulated backscattered data is generated using the script settings and is processed and displayed in real-time at frame rates that are similar to those for live data.

In simulation mode, most GUI processing parameter controls work as before, and this mode permits the post-processing of previously recorded data. The Vantage software can be used without the Vantage hardware system attached, or with the Vantage hardware system attached but powered off. The user can develop scripts in simulation mode and then attach the Vantage hardware system to run the same script in live imaging mode.

In the example above, the "L11-4vFlash" file that was provided as input to VSX is a ".mat" file, which is a MATLAB binary data file. The "SetUpL11_4vFlash.m" MATLAB script, in the "ExampleScripts/L11-4v" subfolder, is the m-script file that generated the "L11-4vFlash.mat" output file. There are many other example scripts illustrating different acquisition geometries and sequences, for B-mode, Color Flow Doppler, Spectral Doppler, unfocused and focused beams, and other acquisition and processing approaches to help with writing customized scripts. Note that an example script does not run if the probe plugged into the system does not match the probe specified in the script, but you can elect to run a script with no probe connected at all. There are exceptions and workarounds: see the [Sequence Programming Manual](#).

If the user attempts to run an example script with a different transducer connected to the system, this results in an error message and the script will not run. If no transducer is connected at all, the user can elect to allow the script to run, for example, to verify that the script and system are functioning, or to examine the transmit output with an oscilloscope, or for other test/ debug purposes.

C.2 Alternative Methods of Running a Script with VSX

As explained in the previous section, the most common way to run a script with VSX is to simply enter VSX at the MATLAB command prompt. This starts VSX and prompts the user for the name of the .mat file that needs to be loaded and executed. The entered name must be identical to the name of the .mat file that was previously created by running a setup script. There are several alternative ways the user can use to run a script with VSX. The user may use any of the alternatives listed below, depending on what you find most convenient:

C.2.1. Using SetUp Script name

When invoking VSX from the MATLAB command prompt as described above and then responding to the query from VSX for the name of the file to use, the user can also enter the name of the SetUp script that was used to create the desired .mat file. After entering a SetUp script name, VSX opens that file and searches within it to find the name of the .mat file the script creates. Subsequently, it uses that .mat file if it exists (note the user still must have run the SetUp script itself at some point in the past, to create the .mat file). This approach saves the user the trouble of having to remember the exact spelling of the .mat file name.

C.2.2. Running VSX automatically from within the SetUp script

A third approach is to add two lines of code within the SetUp script so that when the user runs the script, it automatically runs VSX to load and execute the resulting .mat file. This approach eliminates the need to manually run VSX and also eliminates the need to respond to the query from VSX for the file to be run. To do this, insert the following two lines of code in the setup script immediately after the line where the script saves the .mat file with the "save" command. For example, suppose the user has a SetUp script that creates a file named "myScript.mat". This script would have a save command that looked like this:

```
save('MatFiles/myScript');
```

To automatically run VSX from within the SetUp script, add the following two lines immediately after the line with the save command:

```
filename = 'myScript'; VSX
```

When VSX is invoked, it first searches for a variable named *filename* in the MATLAB workspace. If this variable is found, VSX automatically loads and runs the file identified by *filename* instead of prompting the user to enter it.

Some of the example scripts provided by Verasonics already include those two lines but have them commented out. To enable the automatic run of VSX, uncomment the two lines and then save and run the script.

C.3 Creating Custom Imaging Scripts

The Vantage System is a highly flexible and capable instrument for ultrasound research. You can create your custom imaging scripts to instruct the Vantage hardware system and Vantage software to perform a variety of acquisition sequences. Typically, custom processing algorithms are used with a custom acquisition sequence to produce novel ultrasound displays.

To learn how to write sequence scripts and how to include custom processing functions, first see the [Vantage Sequence Programming Tutorial](#) document. You will also need to refer to the [Vantage Sequence Programming Manual](#) which describes all the Application Programming Interfaces (APIs) and parameter definitions available to the user.

To protect your custom files, you should back up your Vantage project folders regularly.

C.4 Debugging Scripts

The first step in debugging is to examine the contents of the structures created by the Setup script. These are variables in the MATLAB workspace and are examined by double-clicking the variable name in the workspace window. In some cases, VSX creates additional structures and modifies others. The `Version` command is useful to quickly test access to Vantage HAL and hardware.

C.5 Advanced Debugging

Occasionally you may be asked by Verasonics support to acquire a detailed log file to support the investigation of problems in the system software. The system logging is enabled with the following command:

```
vsv.common.util.Logger.start()
```

The logger is stopped with the following command:

```
vsv.common.util.Logger.stop()
```

The log file is created at the root of the software installation. The log file is named using the pattern `VSV_YYYYMMDD_HHMMSS.log`, where `YYYYMMDD_HHMMSS` is the timestamp of the logger start time. Typically, the following sequence of commands will be issued before sending the log file to Verasonics support.

```
vsv.common.util.Logger.start()
```

```
VSX
```

```
<Terminate script execution>
```

```
vsv.common.util.Logger.stop()
```

The log file grows large very quickly so the script should typically only be executed for the minimum amount of time to reproduce the behavior corresponding to the trouble report.